

TOMORROW UNIVERSITY OF APPLIED SCIENCES
MSc in Sustainability, Innovation & Technology
Innovation & Technology Lab – Thesis

Making the Invisible Intelligible

*From Data to Experience: A Machine-Learning Framework for Enhanced
Cognitive Understanding of Complex Climate Tipping Systems*

A Case Study on the Atlantic Meridional Overturning Circulation

Author

Enrico Vaccari

Supervisor

Prof. Jonathan Costa

August 1, 2026

Abstract

[Draft to be written after Discussion is complete.]

Acknowledgements

[To be written. Order: supervisor, family/partner, peers, the RAPID and Copernicus observing communities whose open data made this work possible.]

Contents

Abstract	iii
Acknowledgements	v
List of Abbreviations	xv
I DETECT	1
1 Introduction	3
1.1 The Problem: Cognitive Inaccessibility of Complex Climate Systems	3
1.2 The Gap and the Proposal	3
1.3 The Case Study: AMOC as a Test of the Framework	4
1.4 On the Interdisciplinary Nature of the Work	5
1.5 Contribution, Innovation Relevance and SDG Alignment	5
1.6 Roadmap	6
2 Literature Review	7
2.1 Psychological Barriers to Climate Risk Perception	7
2.2 The Cognitive Science of Dynamic and Complex Systems	8
2.3 Empirical Evaluation of Climate Visualisation	8
2.4 Embodied Cognition and Immersive Representation	10
2.5 Machine Learning as a Representational Substrate	10
2.6 Design Grammar	11
2.7 Synthesis: The Intersection This Thesis Occupies	12
3 Literature Review	15
3.1 Psychological Barriers to Climate Risk Perception	15
3.2 The Cognitive Science of Dynamic and Complex Systems	16
3.3 Empirical Evaluation of Climate Visualisation	16
3.4 Embodied Cognition and Immersive Representation	18
3.5 Machine Learning as a Representational Substrate	18
3.6 Design Grammar	19

3.7	Synthesis: The Intersection This Thesis Occupies	20
4	Background	23
4.1	The Physical System	23
4.2	Bistability and Tipping	23
4.3	Measuring the Circulation	24
4.4	Observational Data	24
5	Methodology	27
5.1	Methodological Approach	27
5.2	Research Design and Workflow	28
5.3	Hypotheses as Methodological Drivers	28
5.4	Data Collection and Analysis Methods	29
5.4.1	The analytical phase	29
5.4.2	The design phase	29
5.4.3	The evaluation phase	30
5.5	Methodological Justification	31
6	The Machine Learning Substrate	33
6.1	Latent Structure & PCA Decomposition	33
6.2	An Essentially Flat Manifold Geometry	35
6.3	Early Warning Signals & Null Result	36
6.4	Anomaly Detection	38
6.5	Observation Against Reanalysis	39
6.6	Analytical Summary	39
II	TRANSLATE	41
7	Designing Experiential Representations	43
7.1	From Latent Structure to Perceptual Form	43
7.2	The Translate Framework	43
7.2.1	Informational Equivalence by Shared Provenance	43
7.2.2	Fidelity	44
7.2.3	Machine Learning as a Substrate	45

7.2.4	The Design Grammar	45
7.2.5	Symmetry of Framing	45
7.3	Three Scenes, One Descent	46
7.3.1	Scene I: The Column	46
7.3.2	Scene II: The Engine Room	48
7.3.3	Scene III: The Threshold	49
III	EVALUATE	53
8	User Study	55
9	Results	57
10	Discussion	59
11	Conclusion and Future Work	61
11.1	Reproducibility and Open Science	61
12	Reflection on Leadership & Personal Growth	63
12.1	Leadership Journey	63
12.2	Inspirational Leadership	63
12.3	Communicating Impactfully	63
12.4	Alignment with Careers Vectors	63
A	Ethics, Boundaries and Limitations	65
A.1	Detect — Analytical Phase	65
A.2	Translate — Design Phase	65
A.3	Evaluate — Empirical Phase	66
A.4	Ethical Conduct of the Study	67
B	Study Protocols	69
B.1	Questionnaire Variants	69
B.2	Structure of the Questionnaire	69
B.3	Comprehension Items	70
B.4	Information Sheet and Consent	70

C Code and Reproducibility	71
C.1 Repositories	71
C.2 Environment and Reproduction	72
C.3 Provenance Ledger	72
References	75

List of Figures

1.1	Sustainable Development Goals addressed by this thesis.	6
2.1	The climate stabilization task	8
2.2	The Cleveland & McGill hierarchy of perceptual tasks.	9
2.3	<i>Artificial Botany</i> by fuse*	12
3.1	The climate stabilization task	16
3.2	The Cleveland & McGill hierarchy of perceptual tasks.	17
3.3	<i>Artificial Botany</i> by fuse*	20
4.1	The Atlantic overturning circulation.	24
6.1	Variance explained by the vertical PCA modes.	34
6.2	The two vertical modes: intensity and redistribution.	35
6.3	The flat latent manifold of AMOC profiles.	36
6.4	Early warning signals: rolling variance and autocorrelation.	37
6.5	Isolation Forest anomalies across the record.	38
7.1	The two conditions of Scene I	46
7.2	The two conditions of Scene II	48
7.3	The two conditions of Scene III	50

List of Tables

2.1	A design grammar for experiential representation of complex systems.	13
3.1	A design grammar for experiential representation of complex systems.	21
5.1	The detect–translate–evaluate workflow.	28
6.1	Variance explained by the vertical PCA modes.	34
6.2	Autoencoder versus PCA at matched latent dimension.	36
6.3	Early-warning trends: naive versus corrected significance.	37
7.1	The three scenes at a glance	51
C.1	Provenance ledger	73

List of Abbreviations

AMOC	Atlantic Meridional Overturning Circulation
EWS	Early Warning Signal
LSTM	Long Short-Term Memory
ML	Machine Learning
PCA	Principal Component Analysis
RAPID	Rapid Climate Change Programme (26°N array)
SDG	Sustainable Development Goals
Sv	Sverdrup ($10^6 \text{ m}^3/\text{s}$)
IF	Isolation Forest

Part I

DETECT

The Context, the Problem, and the Analytical Foundation

...

Chapter 1

Introduction

*Addressed: Background & Context · Personal Mission Alignment · Thesis Statement ·
Innovation Relevance & SDG Alignment*

1.1 The Problem: Cognitive Inaccessibility of Complex Climate Systems

For three decades, climate communication was guided by a simple assumption: the idea that people fail to act because they lack information, and that better facts would close the gap. However, empirical evidence has consistently challenged this assumption. Kellstedt et al. (2008) found that people who knew *more* about climate change often reported *less* concern and a weaker sense of responsibility, even after controlling for politics and demographics. Kahan (2015) showed that scientific literacy can widen rather than close disagreement: the most literate people on opposing sides diverge the most when reading the same evidence. Information, it turns out, is filtered through prior beliefs, cultural identity, and the machinery of cognition. The task is not only to state climate science more accurately, but to present it in a form that matches how people actually make sense of it.

There is a much deeper obstacle that sits beneath this one and predates the climate debate. Across two decades of experiments, Sterman (2008) found that even highly educated people — MIT graduate students, engineers, participants trained in mathematics — reason poorly about systems built on accumulation and flow. Cronin et al. (2009) showed it clearly: given the inflow and outflow of a system and asked to sketch the accumulated stock, many participants confused rates with levels and treated a dynamic process as if it were static. These were not people who lacked the mathematics; they had it, and still the static representation defeated them. The implication for climate is direct. Showing CO₂ emissions and atmospheric concentration as two curves does not convey that the second is the accumulation of the first, so audiences systematically underestimate how much inertia the system carries (Cronin et al., 2009; Sterman, 2008). Cognitive-load theory explains why: working memory is limited, and a representation that asks the viewer to mentally animate a hidden process imposes a burden that grows with the system's complexity (Sweller, 1988).

These two literatures — how climate information is interpreted, and how people reason about dynamic systems — meet at a single conclusion. Communicating a complex climate system is not mainly a problem of accuracy or messaging. It is a problem of *representation*: of finding forms in which the dynamic, multi-scale, potentially non-linear behaviour of such a system becomes something a person can perceive, rather than something they must reconstruct in their head from a static picture.

1.2 The Gap and the Proposal

This representational problem has been approached from two directions, each strong on its own and rarely joined. Climate-visualisation research has documented, in detail, why conventional formats fail non-specialist audiences — and has shown that the visualisations people find clearest are often not the

ones they understand best (Harold et al., 2016; Kause et al., 2021). Separately, machine learning for climate science has grown rapidly, but almost entirely in a *predictive* role: forecasting, detecting anomalies, improving models for expert users (Reichstein et al., 2019; Sonnewald et al., 2021). What has scarcely been tried is using machine learning not to predict, but to *recover the hidden structure* of a climate system and turn it into a form non-specialists can perceive. The gap this thesis occupies is the intersection of the two: the analytical power of the second, directed at the representational problem named by the first.¹

The thesis proposes an integrative framework built in three stages, each drawing on a different tradition and feeding the next. The first is *analytical*: unsupervised machine learning is applied to observational and reanalysis data, not to forecast, but to recover the latent structure of the system — a compact set of coordinates that captures its essential dynamics (Chapter 6). The second is *design-based*: those latent structures are translated into experiential representations, with every design decision grounded in principles from cognitive science, embodied cognition, and information visualisation (Chapter 7). The third is *empirical*: the resulting representations are evaluated against a conventional static baseline through a mixed-methods user study, testing whether they change how non-specialist audiences understand the system (Chapter 8). Put simply: machine learning extracts, design translates, evaluation tests.

1.3 The Case Study: AMOC as a Test of the Framework

A framework like this needs a case demanding enough to test it, and the choice is deliberate: the system should carry the very features that make communication hard.² The Atlantic Meridional Overturning Circulation (AMOC) is such a system. It is invisible in a strong sense: no surface expression reveals it to the eye, and no everyday experience gives access to it. It spans an ocean basin, so no single location shows its behaviour. It varies over decades to centuries, longer than both human experience and the observational record. It is temporally non-linear and a candidate tipping element, whose substantial weakening could shift the circulation into a different regime with hysteresis (Armstrong McKay et al., 2022; Lenton et al., 2008). This matters because a marked slowdown would reorganise North Atlantic heat transport and reshape the climate of Europe (Buckley & Marshall, 2016). Its strength is measured in Sverdrup and observed continuously since 2004 by the RAPID array at 26.5°N (Ben I Moat et al., 2026; Frajka-Williams et al., 2023), with the Copernicus GREP reanalysis extending the record further back.

The AMOC is the system this thesis studies, but not the one most people picture when they think of climate change. When non-specialists are asked to imagine it, they tend to name visible things — melting ice, extreme heat, rising seas (O'Neill & Smith, 2014). That contrast, between climate systems we can see and those that need instruments and computation to perceive at all, is central to the argument here. The two are not independent. The Greenland Ice Sheet, among the most visible signs of a warming world, is physically coupled to the AMOC: freshwater from Greenland lowers the density of North Atlantic surface water, weakens deep-water formation, and can slow the circulation (Böning et al., 2016; Caesar et al., 2018; Rahmstorf et al., 2015). What people can see is linked to what they cannot — and this thesis uses that bridge, from the visible glacier to the hidden current, as the entry point into the invisible system.³

¹ Both literatures are reviewed in full in Chapter 3, which locates the thesis precisely against existing visualisation research, machine-learning applications, and the cognitive-science principles that ground the design.

² The AMOC is used only as a test case. This thesis makes no new oceanographic claim and does not contribute to ocean-circulation research; it engages the AMOC solely through the aspects relevant to the representational problem. Its physical foundations — the governing mechanisms, the Stommel two-box model of bistability, and the formulation of meridional transport — are quickly touched on in Chapter 4 for completeness.

³ The recent behaviour of the AMOC is itself contested: reconstructions suggest long-term weakening and possible approach to

1.4 On the Interdisciplinary Nature of the Work

I began this work driven by a personal mission: *Wisdom Unlocks Awareness, and Awareness Unlocks Agency*. More than a motto, it is the conviction that has guided every decision throughout this journey. I believe that understanding is what gives knowledge the power to move people—not only to know, but to care, and ultimately, to act. People rarely act on a system because they have been told it matters; they act once they grasp it well enough to feel why. This thesis is built at the meeting point of five fields, each carrying part of that problem. Climate science supplies the system, the data, and the standards of evidence. Machine learning supplies the methods that recover hidden structure from data too high-dimensional for the eye — and stands, here, for a wider belief: that emerging technologies, used responsibly and for people rather than instead of them, can become instruments of understanding, not just engines of prediction. Cognitive science explains why dynamic systems defeat intuition, and what lets a mind find its footing. Information visualisation supplies the principles for turning structure into something perceptible. And science communication sets the final test — not whether a representation is accurate, but whether it leaves a non-specialist genuinely understanding, in the mind and in the gut, because a decision made with both is one that holds. The contribution lies in none of these fields alone, but in binding them together. Evidence, the computation that reveals it, and the form in which a person finally meets it are usually kept apart; this thesis refuses that separation — because the future turns on our capacity to understand, together, the systems that will shape it.

1.5 Contribution, Innovation Relevance and SDG Alignment

The contribution is mainly methodological. It is not a new finding about the ocean, nor a new machine-learning technique, but a defensible way of connecting the analytical structure of a complex climate system to the perceptual conditions under which non-specialists meet it — tested against a conventional baseline, so that its value rests on evidence rather than assertion. Positioned against existing work, the approach is distinctive in combining four elements that have not been brought together: recovering a tipping system's latent structure through machine learning, translating it into an experiential representation, grounding each choice in cognitive-science principles, and evaluating the result comparatively.⁴

The relevance runs on three levels. *Academically*, research on communicating complex climate systems has leaned on preference studies — which representation feels clearer — more than on tested comprehension, and experiential formats remain largely unevaluated; this thesis offers a comparative methodology instead of another visualisation. *Institutionally*, it aligns substantively with the United Nations Sustainable Development Goals: primarily **SDG 13 (Climate Action)**, target 13.3 on education, awareness and capacity (*Transforming Our World: The 2030 Agenda for Sustainable Development* | Department of Economic and Social Affairs, n.d.); secondarily **SDG 4 (Quality Education)**, in making complex science accessible; and indirectly **SDG 14 (Life Below Water)**, as the AMOC is part of the ocean system. *Practically*, the framework is built to transfer: the AMOC is one of a family of tipping elements — ice sheets, the Amazon, permafrost, the monsoon — that share invisibility, long timescales, non-linearity and high stakes (Armstrong McKay et al., 2022; Lenton et al., 2008). It is a demanding test

a transition (Boers, 2021; Caesar et al., 2018; Ditlevsen & Ditlevsen, 2023), a reading challenged on methodological grounds (Ben-Yami et al., 2024), while the direct RAPID record shows no clear long-term slowdown (Ben I Moat et al., 2026). This unresolved debate makes clear representation more necessary, not less.

⁴ Four families of related work — conventional static visualisation, interactive climate tools, immersive and experiential formats, and data physicalisation — are examined in Chapter 3. None integrates all four elements above.

case for this matter. Before a society can act on a complex system, it must first be able to picture it; this thesis is one contribution to learning how.



Figure 1.1: The three Sustainable Development Goals this thesis contributes to. SDG 13 (Climate Action), in particular target 13.3 on education, awareness and capacity; SDG 4 (Quality Education), on making complex knowledge accessible; and SDG 14 (Life Below Water), as the AMOC is part of the ocean system. Icons © United Nations. The content of this publication has not been approved by the United Nations and does not reflect its views.

1.6 Roadmap

The thesis follows the detect–translate–evaluate arc in three parts.

Part I — DETECT sets up the problem and the case: Chapter ?? formalises the research questions, objectives and hypotheses; Chapter 3 reviews the relevant literature and locates the thesis within it; Chapter 4 develops the physical and mathematical apparatus needed to treat the AMOC as an analysable object.

Part II — TRANSLATE documents the constructive work: Chapter 5 specifies the three-stage methodology; Chapter 6 presents the analytical pipeline that recovers the system’s latent structure; Chapter 7 designs the representational conditions through which that structure is translated.

Part III — EVALUATE closes the loop: Chapter 8 specifies the user study, Chapter 9 reports its findings, Chapter 10 interprets them, and Chapter 11 reflects on limitations, transferability and future work.

Chapter 2

Literature Review

Addressed: Theoretical Foundations · Current benchmarks · Design Principles · Gaps in the Literature

This chapter reviews the literature on which the thesis is built. Three strands define the problem: the psychology of climate risk perception, the cognitive science of dynamic systems, and the empirical evaluation of climate visualisation. Two supply the response: embodied and immersive representation, and machine learning used representationally rather than predictively. A sixth strand translates the theory into a design grammar for building an experiential representation, and a closing section names the gap the thesis occupies.

2.1 Psychological Barriers to Climate Risk Perception

For much of the past three decades, climate communication rested on what is called the *information-deficit hypothesis*: the belief that inaction reflected missing information, and that closing the knowledge gap would close the action gap. The evidence has clearly challenged this assumption. Kellstedt et al. (2008) found that people with higher climate knowledge reported *lower* concern and *lower* sense of responsibility.¹ Kahan (2015) showed that scientific literacy *polarised* climate attitudes rather than converging them, so the most literate members of opposing groups disagreed most. The lesson is not that information does not matter, but that its effect is mediated by cognitive and cultural machinery the standard formats never addressed.

The most influential account of this failure is Stoknes (2015), whose five barriers — distance, doom, dissonance, denial and identity — have organised much later work.² The barriers interlock rather than act singly (Van Der Linden, 2015), and audiences stratify into psychologically distinct groups whose barriers activate differently, so no single strategy serves all (*Global Warming's Six Americas, September 2021*, n.d.). Conventional visualisation has not dissolved them, and the reason is structural. The imagery most used to communicate climate change either reinforces the very distance communicators hope to close or triggers the doom response that suppresses engagement (Creative, n.d.; Nicholson-Cole, 2005). Formats that give the reader no active role consistently underperform those that do (Ojala, 2023). Adding precision to a line graph opens no such role: its implicit rhetoric of *here is the fact* leaves the reader nowhere to stand. The recent literature therefore points toward representations that treat the reader as an active participant in making meaning (Marlon et al., 2019; Morris et al., 2019) — the direction this thesis pursues, with machine learning as its substrate.

¹ The inverse relationship survived controls for political affiliation and demographics, indicating a genuine cognitive-cultural effect rather than a confound.

² Briefly: *distance* (climate change feels spatially, temporally and socially remote), *doom* (catastrophist framing triggers avoidance), *dissonance* (the belief-behaviour gap is resolved by adjusting the belief), *denial* (self-protective disengagement) and *identity* (climate positions mark group belonging).

2.2 The Cognitive Science of Dynamic and Complex Systems

Beneath these cultural filters lies a barrier independent of them: the mind's structural limits in reasoning about systems of accumulation, feedback and delay. These are stable features of cognitive architecture, not gaps better teaching could close, and any representation that ignores them will fail however well it is tuned psychologically.

The strongest evidence is Sterman (2008), whose experiments show that even mathematically literate adults make systematic errors on elementary stock-and-flow problems. Shown a scenario in which atmospheric CO₂ stabilises, and asked to sketch the emissions path consistent with it, most violate mass balance — matching the shape of the output to the shape of the input rather than reasoning about the difference between inflow and outflow (Figures ??–??).³ Cronin et al. (2009) found the same in starker form: asked to sketch a stock from its inflow and outflow, participants confused rates with levels and treated the dynamic system as static, even when they had the mathematical apparatus to solve it. Given only a graph, the mind does not spontaneously build the dynamic scenario the graph depicts.

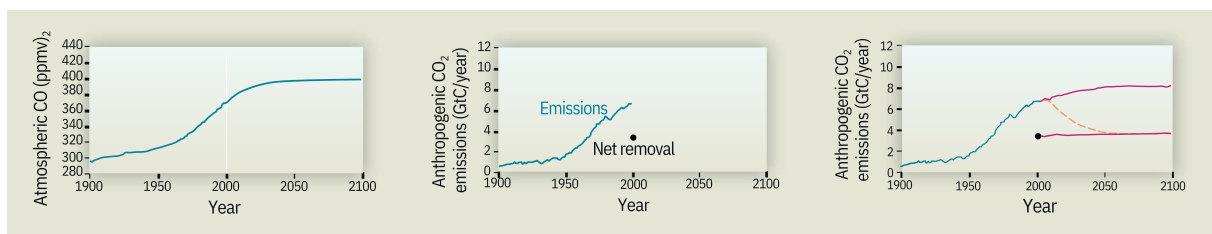


Figure 2.1: The climate stabilization task. (a) The scenario presented to participants: atmospheric CO₂ concentration rises gradually to 400 ppmv and stabilizes by 2100. (b) The stock–flow information provided: anthropogenic CO₂ emissions from 1900 to 2000 (the inflow) and current net removal by oceans and biomass (the outflow), with emissions roughly double net removal. (c) A typical response: participants erroneously sketch future emissions that match the shape of the CO₂ trajectory (pattern-matching heuristic), while mass balance requires emissions to fall until they equal net removal (dashed line). Reprinted from Sterman (2008).

Cognitive-load theory explains why (Sweller, 1988; Sweller et al., 2019). Working memory has strictly limited capacity, and a static image of a dynamic system asks the reader to mentally animate what the image does not animate — a load imposed on top of the domain itself. Ware (2021) argues that a communicative visualisation must lower this internal-simulation demand by making dynamic structure perceptually available in the representation itself. A third strand points ahead: B. G. Tversky (2019), synthesising four decades of work, shows that human reasoning about complex processes is fundamentally spatial, understood through bodily and kinaesthetic schemas the mind already inhabits. If the AMOC is structurally dynamic and human reasoning about dynamics is spatially embodied, the strategies most likely to succeed are those that enrol the reader's spatial and kinaesthetic cognition. That proposition organises the rest of the chapter.

2.3 Empirical Evaluation of Climate Visualisation

What has actually been measured about how climate visualisations are received reveals a methodological asymmetry: the field is rich in studies of what audiences *prefer* and sparse in studies of what they *understand*. That asymmetry defines the gap this thesis addresses. The most systematic review found

³ In Sterman's studies 84% of highly educated participants — MIT graduate students, engineers, participants trained in calculus — made this error, indicating a limit of representation rather than of arithmetic.

that standard formats — line graphs, choropleth maps, ensemble spaghetti plots — are mismatched to their audiences’ cognitive resources, and that the mismatch appears even among trained policy readers (Harold et al., 2016). Those readers underestimate uncertainty ranges shown as shaded envelopes and misread ensemble spread as disagreement rather than as a structural feature of a projection (Lang, 2014). Most tellingly, the visualisations people *prefer* are not those that produce the highest *comprehension*: people prefer what feels familiar and understand what is structurally aligned with the phenomenon (Kause et al., 2020).⁴ Optimising for preference therefore risks producing what Sheppard (2015) calls *consoling visualisations* — images received warmly and understood poorly.

This is not accidental; it rests on a mature science of visual encoding that Chapter 7 draws on directly. Bertin identified the finite set of *visual variables* through which any mark encodes data — position, size, value, colour, orientation, texture (Bertin et al., 1983). Cleveland and McGill (1984) ranked them by decoding accuracy: position along a common scale is judged far more accurately than length, angle, area or colour saturation. Sacha et al. (2017a) turned this into a design rule, *effectiveness*: match the most important information to the most accurately decoded channel. Ware (2021) grounds the same rule in perception, distinguishing channels processed pre-attentively — in parallel, before focused attention — from those needing serial inspection. Encoding, in short, is measured perceptual performance, not taste.

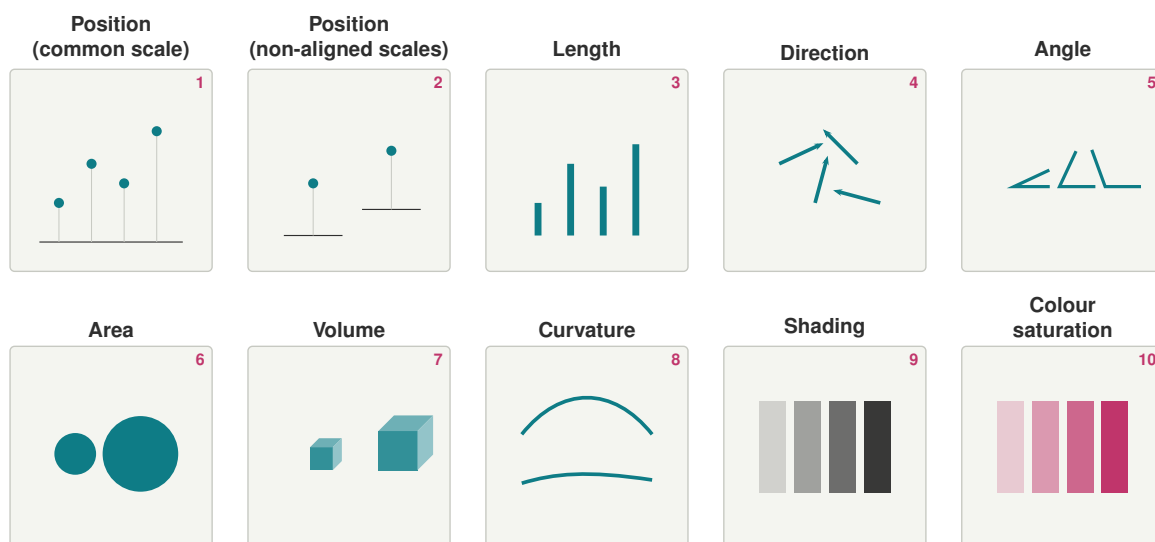


Figure 2.2: The hierarchy of elementary perceptual tasks, after Cleveland and McGill (1984). Encodings are ordered by decoding accuracy, from most accurate (1, position on a common scale) to least accurate (10, colour saturation). The ranking is empirical, not aesthetic: it reflects how accurately people extract a quantitative value from each visual channel. It underlies design-grammar principle P0 (Table 3.1) — assign the most important variable to the most accurately decoded channel — and principle P1, which places continuous magnitude on spatial position.

Uncertainty is where conventional encoding fails most, and where this thesis departs most from convention. Error bars and shaded bands are read poorly by non-specialists, who treat a band as a hard edge (Spiegelhalter, 2017); Padilla et al. (2018) explain why, showing that static uncertainty encodings overload working memory, so viewers fall back on heuristics that discard the uncertainty. The best-evidenced alternative is to *animate* it: Hypothetical Outcome Plots show a sequence of possible outcomes over time, letting the viewer feel the spread instead of decoding a band, and untrained observers judge them more

⁴ A later study found the same split for uncertainty specifically: intuitive icon arrays produced overconfident readings, while less intuitive but structurally faithful frequency formats produced calibrated ones (Kause et al., 2021).

accurately than error bars (Hullman et al., 2015; Kale et al., 2019). This licenses the thesis’s central move — rendering uncertainty as a felt perceptual quality rather than a bracketed number.

These limits are visible in how the AMOC itself is communicated, which fixes the study’s control condition. In the scientific and public record the AMOC appears almost entirely as line graphs of transport in Sverdrup, model-ensemble projections of decline shown as a shaded spread, and reconstructed “fingerprint” indices, with the IPCC adding verbal confidence qualifiers such as “very likely to weaken” (ipcc2021; Caesar et al., 2018). Each form is limited exactly as the sections above predict: the static curve depicts a dynamic system in its final state; the shaded band encodes uncertainty in the way shown to be misread; the verbal qualifiers ask a non-specialist to hold a probabilistic threshold judgement in working memory with no perceptual support; and none conveys the non-linearity that defines a tipping element. The conventional baseline communicates the AMOC’s magnitude and trend while failing to communicate its dynamics, non-linearity and uncertainty — precisely what the experiential condition is designed to improve upon, and reconstructing it faithfully is what makes the comparison in Chapter 8 fair.

2.4 Embodied Cognition and Immersive Representation

If the response is to enrol spatial and bodily cognition, three related traditions show how. The philosophical grounding is Lakoff (1999): the abstract concepts through which we reason about systems — acceleration, feedback, stability, tipping — are extensions of bodily and spatial experience, not disembodied logic. Kirsh (2013) supplied experimental support, finding that people allowed to gesture and move while reasoning about complex processes outperformed those restricted to static inspection. The most concrete design tradition to grow from this is *data physicalisation* (Jansen et al., 2015), whose artefacts encode information for exploration through touch and sight; participants working with them recall represented information more accurately than from equivalent charts (“[Cover Art]”, 2009). Parallel to it, *data humanism* argues that data communication has been dominated by an aesthetic of dispassionate abstraction, and that narrative, authored representation engages readers more deeply (PrintMag, 2017; Segel & Heer, 2010). The third tradition, *immersive analytics*, holds that immersive and spatially-aware interfaces offer representational affordances the flat screen cannot, suited to data whose structure is spatial or dynamic (Marriott et al., 2018).⁵ Browser-based rendering and real-time audio now bring these affordances within reach of a single research project, which is the opening this thesis takes up. Together these traditions support one claim: the difficulty of communicating complex climate systems is not addressable by refinement within the conventional paradigm, and the literatures point jointly toward treating the reader as an active, embodied participant.

2.5 Machine Learning as a Representational Substrate

Machine learning for climate science is now vast, but its dominant agenda is *predictive*: forecasting variables, detecting anomalies, parameterising physical processes, downscaling model output (Reichstein et al., 2019; Sonnewald et al., 2021). The AMOC literature mirrors this, applying machine learning mostly to detect early-warning signals or forecast variability (Boers, 2021; Ditlevsen & Ditlevsen, 2023). This work is valuable, but it is oriented almost entirely toward new results for expert audiences, leaving an

⁵ Formal evaluations of immersive analytics remain limited, and the strongest current evidence concerns three-dimensional spatial and temporal tasks rather than abstract complex systems. The theoretical alignment with the cognitive science of §3.2 is nonetheless direct.

adjacent question unexplored: what machine learning contributes when the goal is not a new result but to make an existing scientific object *intelligible to non-specialists*.

The distinction is worth stating precisely. A network forecasting AMOC transport produces a new capability, judged in shared units — correlation, error, calibration. A dimensionality-reduction analysis of the same record produces a latent representation whose value lies elsewhere: in what few coordinates account for most of the variability, and what they mean physically. The thesis uses machine learning in this second, *representational* role. The claim is not that the AMOC’s vertical structure is novel — it is consistent with decades of oceanography (Buckley & Marshall, 2016; Hannachi et al., 2007) — but that this structure, once recovered, can become the coordinate system through which the AMOC is made available to perception. The literature treating machine learning this way is thin but real: latent representations have been argued to be legitimate epistemological objects in physics (Iten et al., 2020), and visual-analytics research treats a model’s latent space as the substrate through which an analyst builds understanding (Sacha et al., 2017b). Extending this from expert analytics to public communication of a climate system has been proposed occasionally (Cabrera et al., 2023) but scarcely implemented. That is the gap.

2.6 Design Grammar

The previous sections show why conventional visualisation often fails and why an embodied approach offers an alternative. They do not yet explain *how* such an approach should be designed. This section closes that gap by deriving design principles from the literature above: each links a system property to a perceptual channel, supported by evidence from cognitive science and visualisation research, so that the choices in Chapter 7 are informed design decisions rather than aesthetic ones — each traceable to a foundation and defensible on its own.

The parent principle is the effectiveness rule established in §3.3: the most important information should be assigned to the most accurately decoded visual channel (Bertin et al., 1983; Cleveland & McGill, 1984; Sacha et al., 2017a). The remaining principles follow by applying it to the characteristics of complex systems: magnitude through spatial position, change over time through motion, anomalies through density or texture, uncertainty through controlled variation rather than static intervals, and secondary ordered information through sound. These mappings rest on the evidence discussed above and are stated as general principles applicable to invisible systems beyond the AMOC. Table 3.1 summarises the grammar; Chapter 7 applies it to the system properties recovered in Chapter 6.

The grammar also positions the thesis within the broader field of experiential data representation. Data-art practice already explores how hidden structures from computational models can become perceptual experiences: the studio fuse*, for example, treats the latent spaces of generative models as environments of continuous transformation.⁶ These approaches share an assumption with this thesis — that computational structure can become something to experience, not only to analyse — but they do not test whether an audience *understands* anything as a result. The contribution here is the step that practice omits: to ground each perceptual choice in the principles of Table 3.1, and to test comparatively whether the result supports both meaningful experience and measurable understanding.

⁶ fuse*, *Artificial Botany* and related works, <https://fuseworks.it>. The studio describes its interest as the “intermediate dimension” of latent space, explored as a space of expressive exploration rather than as a direct representation of a measured system.



Figure 2.3: Artificial Botany by fuse*. The work explores the latent space of generative networks as a terrain of morphing forms. Description adapted from “Artificial Botany” (2019–present).

2.7 Synthesis: The Intersection This Thesis Occupies

Read as one argument, the strands converge on a proposition none states alone. Climate information is filtered through psychological barriers that conventional formats fail to address (§3.1); those barriers are compounded by structural limits on reasoning about dynamics from static images (§3.2); climate visualisation has been evaluated mostly for preference rather than comprehension, though a mature science of encoding exists to ground better choices (§3.3); embodied and immersive traditions show the direction of a response but have rarely been applied to climate communication (§3.4); and machine learning has advanced predictively, leaving the representational role of latent structure almost unexplored (§3.5). The design grammar (§3.6) shows these strands can be combined into operational principles — but no existing work has done so and tested the result.

The intersection is narrow and empirically unpopulated, defined by five commitments: a climate system whose invisibility and non-linearity make it a stress test (Armstrong McKay et al., 2022; Lenton et al., 2008); a machine-learning treatment that recovers latent structure rather than forecasts (Iten et al., 2020; Sonnewald et al., 2019); a design tradition that treats the reader as embodied and active (Jansen et al., 2015; Lakoff, 1999; Marriott et al., 2018); a comparison of conventional and experiential formats (Harold et al., 2016; Kause et al., 2020); and a qualitative, exploratory evaluation appropriate to a phenomenon not yet ripe for population-level inference (Braun & Clarke, 2006). No study reviewed satisfies all five jointly. The gap is not the absence of any one commitment but the absence of their integration — and integrating them, on the AMOC as a case, is what the chapters that follow undertake.

Table 2.1: A design grammar for experiential representation of complex systems. Each row maps a *system property* to a *perceptual channel* through a documented *cognitive principle*, anchored to a *source*. P0 is the parent principle — decoding accuracy is ranked, not uniform — from which P1–P5 follow by application to the properties of a complex system. The grammar is stated in general, transferable form here; its application to the specific AMOC features recovered in Chapter 6 (latent intensity, transitions, anomalies, ensemble uncertainty) is developed in Chapter 7. Motion (P2), animated uncertainty (P4) and sound (P5) are the channels through which this grammar goes beyond conventional static visualisation.

	System property	Perceptual channel	Cognitive principle	Source
P0	Relative importance of variables	Most accurate channel for the most important variable	Effectiveness: decoding accuracy is ranked, not uniform	Bertin et al. (1983), Cleveland & McGill (1984), and Sacha et al. (2017a)
P1	Magnitude / intensity (ordered, continuous)	Spatial position on a common axis	Position is the most accurately decoded channel for ordered data	Cleveland & McGill (1984) and B. G. Tversky (2019)
P2	Dynamics / change over time	Actual motion	The mind cannot mentally animate a static frame; motion is tracked pre-attentively	Cronin et al. (2009), Sterman (2008), and Ware (2021)
P3	Anomaly / departure from regime	Local density or texture change	Irregularity draws bottom-up attention without a legend	Bertin et al. (1983) and Ware (2021)
P4	Uncertainty	Animated variability (motion, density, sound), not a static band	Static bands overload working memory; animated outcomes are judged more accurately	Hullman et al. (2015), Kale et al. (2019), and Padilla et al. (2018)
P5	Secondary or-dered variable	Sound (pitch, rhythm, timbre)	Auditory channels carry ordered data in parallel, without visual clutter	<i>The Sonification Handbook</i> Edited by Hermann, Hunt, Neuhoﬀ (n.d.)

Chapter 3

Literature Review

Addressed: Theoretical Foundations · Current benchmarks · Design Principles · Gaps in the Literature

This chapter reviews the literature on which the thesis is built. Three strands define the problem: the psychology of climate risk perception, the cognitive science of dynamic systems, and the empirical evaluation of climate visualisation. Two supply the response: embodied and immersive representation, and machine learning used representationally rather than predictively. A sixth strand translates the theory into a design grammar for building an experiential representation, and a closing section names the gap the thesis occupies.

3.1 Psychological Barriers to Climate Risk Perception

As mentioned earlier, for much of the past three decades, climate communication rested on what is called the *information-deficit hypothesis*: the belief that inaction reflected missing information, and that closing the knowledge gap would close the action gap. The evidence has clearly challenged this assumption. Kellstedt et al. (2008) found that people with higher climate knowledge reported *lower* concern and *lower* sense of responsibility.¹ Kahan (2015) showed that scientific literacy *polarised* climate attitudes rather than converging them, so the most literate members of opposing groups disagreed most. The lesson is not that information does not matter, but that its effect is mediated by cognitive and cultural machinery the standard formats never addressed.

The most influential account of this failure is Stoknes (2015), whose five barriers — distance, doom, dissonance, denial and identity — have organised much later work.² The barriers interlock rather than act singly (Van Der Linden, 2015), and audiences stratify into psychologically distinct groups whose barriers activate differently, so no single strategy serves all (*Global Warming's Six Americas, September 2021*, n.d.). Conventional visualisation has not dissolved them, and the reason is structural. The imagery most used to communicate climate change either reinforces the very distance communicators hope to close or triggers the doom response that suppresses engagement (Creative, n.d.; Nicholson-Cole, 2005). Formats that give the reader no active role consistently underperform those that do (Ojala, 2023). Adding precision to a line graph opens no such role: its implicit rhetoric of *here is the fact* leaves the reader nowhere to stand. The recent literature therefore points toward representations that treat the reader as an active participant in making meaning (Marlon et al., 2019; Morris et al., 2019) — the direction this thesis pursues, with machine learning as its substrate.

¹ The inverse relationship survived controls for political affiliation and demographics, indicating a genuine cognitive-cultural effect rather than a confound.

² Briefly: *distance* (climate change feels spatially, temporally and socially remote), *doom* (catastrophist framing triggers avoidance), *dissonance* (the belief-behaviour gap is resolved by adjusting the belief), *denial* (self-protective disengagement) and *identity* (climate positions mark group belonging).

3.2 The Cognitive Science of Dynamic and Complex Systems

Beneath these cultural filters lies a barrier independent of them: the mind's structural limits in reasoning about systems of accumulation, feedback and delay. These are stable features of cognitive architecture, not gaps better teaching could close, and any representation that ignores them will fail however well it is tuned psychologically.

The strongest evidence is Sterman (2008), whose experiments show that even mathematically literate adults make systematic errors on elementary stock-and-flow problems. Shown a scenario in which atmospheric CO₂ stabilises, and asked to sketch the emissions path consistent with it, most violate mass balance — matching the shape of the output to the shape of the input rather than reasoning about the difference between inflow and outflow (Figures ??–??).³ Cronin et al. (2009) found the same in starker form: asked to sketch a stock from its inflow and outflow, participants confused rates with levels and treated the dynamic system as static, even when they had the mathematical apparatus to solve it. Given only a graph, the mind does not spontaneously build the dynamic scenario the graph depicts.

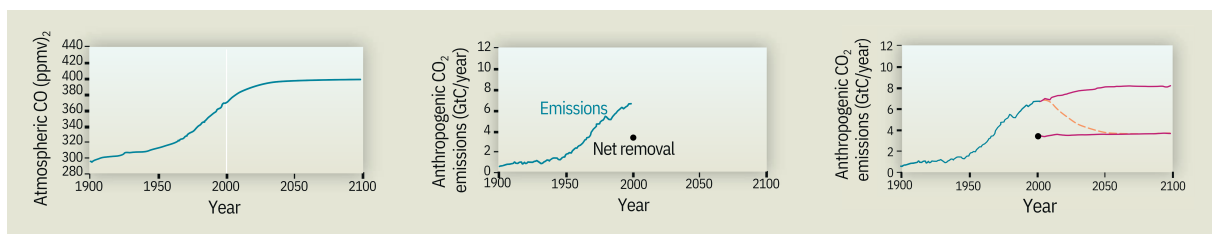


Figure 3.1: The climate stabilization task. (a) The scenario presented to participants: atmospheric CO₂ concentration rises gradually to 400 ppmv and stabilizes by 2100. (b) The stock–flow information provided: anthropogenic CO₂ emissions from 1900 to 2000 (the inflow) and current net removal by oceans and biomass (the outflow), with emissions roughly double net removal. (c) A typical response: participants erroneously sketch future emissions that match the shape of the CO₂ trajectory (pattern-matching heuristic), while mass balance requires emissions to fall until they equal net removal (dashed line). Reprinted from Sterman (2008).

Cognitive-load theory explains why (Sweller, 1988; Sweller et al., 2019). Working memory has strictly limited capacity, and a static image of a dynamic system asks the reader to mentally animate what the image does not animate — a load imposed on top of the domain itself. Ware (2021) argues that a communicative visualisation must lower this internal-simulation demand by making dynamic structure perceptually available in the representation itself. A third strand points ahead: B. G. Tversky (2019), synthesising four decades of work, shows that human reasoning about complex processes is fundamentally spatial, understood through bodily and kinaesthetic schemas the mind already inhabits. If the AMOC is structurally dynamic and human reasoning about dynamics is spatially embodied, the strategies most likely to succeed are those that enrol the reader's spatial and kinaesthetic cognition. That proposition organises the rest of the chapter.

3.3 Empirical Evaluation of Climate Visualisation

What has actually been measured about how climate visualisations are received reveals a methodological asymmetry: the field is rich in studies of what audiences *prefer* and sparse in studies of what they *understand*. That asymmetry defines the gap this thesis addresses. The most systematic review found

³ In Sterman's studies 84% of highly educated participants — MIT graduate students, engineers, participants trained in calculus — made this error, indicating a limit of representation rather than of arithmetic.

that standard formats — line graphs, choropleth maps, ensemble spaghetti plots — are mismatched to their audiences’ cognitive resources, and that the mismatch appears even among trained policy readers (Harold et al., 2016). Those readers underestimate uncertainty ranges shown as shaded envelopes and misread ensemble spread as disagreement rather than as a structural feature of a projection (Lang, 2014). Most tellingly, the visualisations people *prefer* are not those that produce the highest *comprehension*: people prefer what feels familiar and understand what is structurally aligned with the phenomenon (Kause et al., 2020).⁴ Optimising for preference therefore risks producing what Sheppard (2015) calls *consoling visualisations* — images received warmly and understood poorly.

This is not accidental; it rests on a mature science of visual encoding that Chapter 7 draws on directly. Bertin identified the finite set of *visual variables* through which any mark encodes data — position, size, value, colour, orientation, texture (Bertin et al., 1983). Cleveland and McGill (1984) ranked them by decoding accuracy: position along a common scale is judged far more accurately than length, angle, area or colour saturation. Sacha et al. (2017a) turned this into a design rule, *effectiveness*: match the most important information to the most accurately decoded channel. Ware (2021) grounds the same rule in perception, distinguishing channels processed pre-attentively — in parallel, before focused attention — from those needing serial inspection. Encoding, in short, is measured perceptual performance, not taste.

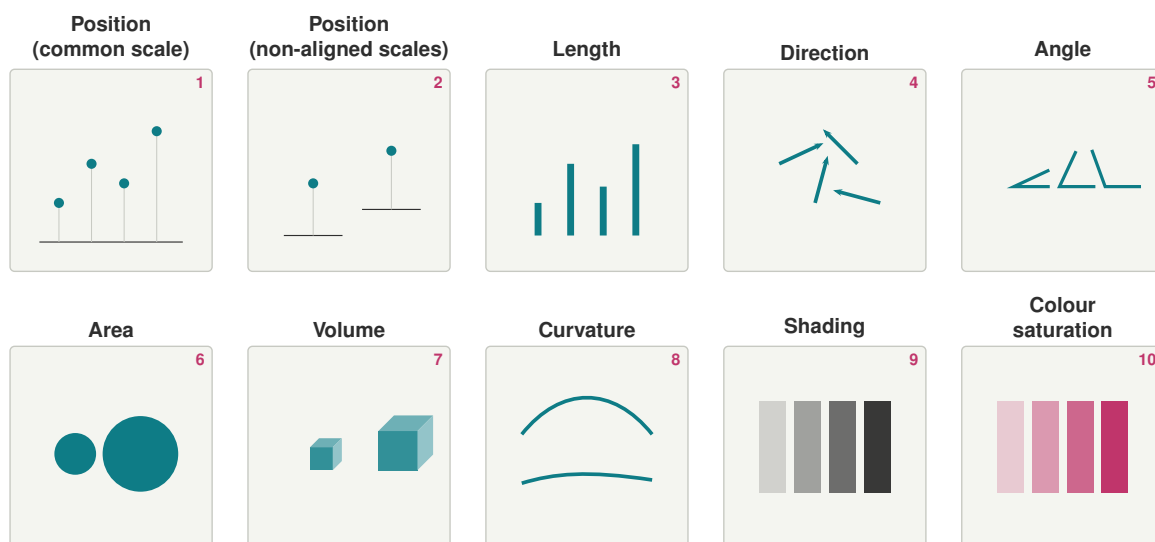


Figure 3.2: The hierarchy of elementary perceptual tasks, after Cleveland and McGill (1984). Encodings are ordered by decoding accuracy, from most accurate (1, position on a common scale) to least accurate (10, colour saturation). The ranking is empirical, not aesthetic: it reflects how accurately people extract a quantitative value from each visual channel. It underlies design-grammar principle P0 (Table 3.1) — assign the most important variable to the most accurately decoded channel — and principle P1, which places continuous magnitude on spatial position.

Uncertainty is where conventional encoding fails most, and where this thesis departs most from convention. Error bars and shaded bands are read poorly by non-specialists, who treat a band as a hard edge (Spiegelhalter, 2017); Padilla et al. (2018) explain why, showing that static uncertainty encodings overload working memory, so viewers fall back on heuristics that discard the uncertainty. The best-evidenced alternative is to *animate* it: Hypothetical Outcome Plots show a sequence of possible outcomes over time, letting the viewer feel the spread instead of decoding a band, and untrained observers judge them more

⁴ A later study found the same split for uncertainty specifically: intuitive icon arrays produced overconfident readings, while less intuitive but structurally faithful frequency formats produced calibrated ones (Kause et al., 2021).

accurately than error bars (Hullman et al., 2015; Kale et al., 2019). This licenses the thesis’s central move — rendering uncertainty as a felt perceptual quality rather than a bracketed number.

These limits are visible in how the AMOC itself is communicated, which fixes the study’s control condition. In the scientific and public record the AMOC appears almost entirely as line graphs of transport in Sverdrup, model-ensemble projections of decline shown as a shaded spread, and reconstructed “fingerprint” indices, with the IPCC adding verbal confidence qualifiers such as “very likely to weaken” (ipcc2021; Caesar et al., 2018). Each form is limited exactly as the sections above predict: the static curve depicts a dynamic system in its final state; the shaded band encodes uncertainty in the way shown to be misread; the verbal qualifiers ask a non-specialist to hold a probabilistic threshold judgement in working memory with no perceptual support; and none conveys the non-linearity that defines a tipping element. The conventional baseline communicates the AMOC’s magnitude and trend while failing to communicate its dynamics, non-linearity and uncertainty — precisely what the experiential condition is designed to improve upon, and reconstructing it faithfully is what makes the comparison in Chapter 8 fair.

3.4 Embodied Cognition and Immersive Representation

If the response is to enrol spatial and bodily cognition, three related traditions show how. The philosophical grounding is Lakoff (1999): the abstract concepts through which we reason about systems — acceleration, feedback, stability, tipping — are extensions of bodily and spatial experience, not disembodied logic. Kirsh (2013) supplied experimental support, finding that people allowed to gesture and move while reasoning about complex processes outperformed those restricted to static inspection. The most concrete design tradition to grow from this is *data physicalisation* (Jansen et al., 2015), whose artefacts encode information for exploration through touch and sight; participants working with them recall represented information more accurately than from equivalent charts (“[Cover Art]”, 2009). Parallel to it, *data humanism* argues that data communication has been dominated by an aesthetic of dispassionate abstraction, and that narrative, authored representation engages readers more deeply (PrintMag, 2017; Segel & Heer, 2010). The third tradition, *immersive analytics*, holds that immersive and spatially-aware interfaces offer representational affordances the flat screen cannot, suited to data whose structure is spatial or dynamic (Marriott et al., 2018).⁵ Browser-based rendering and real-time audio now bring these affordances within reach of a single research project, which is the opening this thesis takes up. Together these traditions support one claim: the difficulty of communicating complex climate systems is not addressable by refinement within the conventional paradigm, and the literatures point jointly toward treating the reader as an active, embodied participant.

3.5 Machine Learning as a Representational Substrate

Machine learning for climate science is now vast, but its dominant agenda is *predictive*: forecasting variables, detecting anomalies, parameterising physical processes, downscaling model output (Reichstein et al., 2019; Sonnewald et al., 2021). The AMOC literature mirrors this, applying machine learning mostly to detect early-warning signals or forecast variability (Boers, 2021; Ditlevsen & Ditlevsen, 2023). This work is valuable, but it is oriented almost entirely toward new results for expert audiences, leaving an

⁵ Formal evaluations of immersive analytics remain limited, and the strongest current evidence concerns three-dimensional spatial and temporal tasks rather than abstract complex systems. The theoretical alignment with the cognitive science of §3.2 is nonetheless direct.

adjacent question unexplored: what machine learning contributes when the goal is not a new result but to make an existing scientific object *intelligible to non-specialists*.

The distinction is worth stating precisely. A network forecasting AMOC transport produces a new capability, judged in shared units — correlation, error, calibration. A dimensionality-reduction analysis of the same record produces a latent representation whose value lies elsewhere: in what few coordinates account for most of the variability, and what they mean physically. The thesis uses machine learning in this second, *representational* role. The claim is not that the AMOC’s vertical structure is novel — it is consistent with decades of oceanography (Buckley & Marshall, 2016; Hannachi et al., 2007) — but that this structure, once recovered, can become the coordinate system through which the AMOC is made available to perception. The literature treating machine learning this way is thin but real: latent representations have been argued to be legitimate epistemological objects in physics (Iten et al., 2020), and visual-analytics research treats a model’s latent space as the substrate through which an analyst builds understanding (Sacha et al., 2017b). Extending this from expert analytics to public communication of a climate system has been proposed occasionally (Cabrera et al., 2023) but scarcely implemented. That is the gap.

3.6 Design Grammar

The previous sections show why conventional visualisation often fails and why an embodied approach offers an alternative. They do not yet explain *how* such an approach should be designed. This section closes that gap by deriving design principles from the literature above: each links a system property to a perceptual channel, supported by evidence from cognitive science and visualisation research, so that the choices in Chapter 7 are informed design decisions rather than aesthetic ones — each traceable to a foundation and defensible on its own.

The parent principle is the effectiveness rule established in §3.3: the most important information should be assigned to the most accurately decoded visual channel (Bertin et al., 1983; Cleveland & McGill, 1984; Sacha et al., 2017a). The remaining principles follow by applying it to the characteristics of complex systems: magnitude through spatial position, change over time through motion, anomalies through density or texture, uncertainty through controlled variation rather than static intervals, and secondary ordered information through sound. These mappings rest on the evidence discussed above and are stated as general principles applicable to invisible systems beyond the AMOC. Table 3.1 summarises the grammar; Chapter 7 applies it to the system properties recovered in Chapter 6.

The grammar also positions the thesis within the broader field of experiential data representation. Data-art practice already explores how hidden structures from computational models can become perceptual experiences: the studio fuse*, for example, treats the latent spaces of generative models as environments of continuous transformation.⁶ These approaches share an assumption with this thesis — that computational structure can become something to experience, not only to analyse — but they do not test whether an audience *understands* anything as a result. The contribution here is the step that practice omits: to ground each perceptual choice in the principles of Table 3.1, and to test comparatively whether the result supports both meaningful experience and measurable understanding.

⁶ fuse*, *Artificial Botany* and related works, <https://fuseworks.it>. The studio describes its interest as the “intermediate dimension” of latent space, explored as a space of expressive exploration rather than as a direct representation of a measured system.



Figure 3.3: Artificial Botany by fuse*. The work explores the latent space of generative networks as a terrain of morphing forms. Description adapted from “Artificial Botany” (2019–present).

3.7 Synthesis: The Intersection This Thesis Occupies

Read as one argument, the strands converge on a proposition none states alone. Climate information is filtered through psychological barriers that conventional formats fail to address (§3.1); those barriers are compounded by structural limits on reasoning about dynamics from static images (§3.2); climate visualisation has been evaluated mostly for preference rather than comprehension, though a mature science of encoding exists to ground better choices (§3.3); embodied and immersive traditions show the direction of a response but have rarely been applied to climate communication (§3.4); and machine learning has advanced predictively, leaving the representational role of latent structure almost unexplored (§3.5). The design grammar (§3.6) shows these strands can be combined into operational principles — but no existing work has done so and tested the result.

The intersection is narrow and empirically unpopulated, defined by five commitments: a climate system whose invisibility and non-linearity make it a stress test (Armstrong McKay et al., 2022; Lenton et al., 2008); a machine-learning treatment that recovers latent structure rather than forecasts (Iten et al., 2020; Sonnewald et al., 2019); a design tradition that treats the reader as embodied and active (Jansen et al., 2015; Lakoff, 1999; Marriott et al., 2018); a comparison of conventional and experiential formats (Harold et al., 2016; Kause et al., 2020); and a qualitative, exploratory evaluation appropriate to a phenomenon not yet ripe for population-level inference (Braun & Clarke, 2006). No study reviewed satisfies all five jointly. The gap is not the absence of any one commitment but the absence of their integration — and integrating them, on the AMOC as a case, is what the chapters that follow undertake.

Table 3.1: A design grammar for experiential representation of complex systems. Each row maps a *system property* to a *perceptual channel* through a documented *cognitive principle*, anchored to a *source*. P0 is the parent principle — decoding accuracy is ranked, not uniform — from which P1–P5 follow by application to the properties of a complex system. The grammar is stated in general, transferable form here; its application to the specific AMOC features recovered in Chapter 6 (latent intensity, transitions, anomalies, ensemble uncertainty) is developed in Chapter 7. Motion (P2), animated uncertainty (P4) and sound (P5) are the channels through which this grammar goes beyond conventional static visualisation.

	System property	Perceptual channel	Cognitive principle	Source
P0	Relative importance of variables	Most accurate channel for the most important variable	Effectiveness: decoding accuracy is ranked, not uniform	Bertin et al. (1983), Cleveland & McGill (1984), and Sacha et al. (2017a)
P1	Magnitude / intensity (ordered, continuous)	Spatial position on a common axis	Position is the most accurately decoded channel for ordered data	Cleveland & McGill (1984) and B. G. Tversky (2019)
P2	Dynamics / change over time	Actual motion	The mind cannot mentally animate a static frame; motion is tracked pre-attentively	Cronin et al. (2009), Sterman (2008), and Ware (2021)
P3	Anomaly / departure from regime	Local density or texture change	Irregularity draws bottom-up attention without a legend	Bertin et al. (1983) and Ware (2021)
P4	Uncertainty	Animated variability (motion, density, sound), not a static band	Static bands overload working memory; animated outcomes are judged more accurately	Hullman et al. (2015), Kale et al. (2019), and Padilla et al. (2018)
P5	Secondary ordered variable	Sound (pitch, rhythm, timbre)	Auditory channels carry ordered data in parallel, without visual clutter	<i>The Sonification Handbook</i> Edited by Hermann, Hunt, Neuhoﬀ (n.d.)

Chapter 4

Background

Addressed: Physical System · Bistability & Tipping · Measurement · Observational Data

This chapter sets out what the AMOC is as a physical, measurable object — the apparatus the later chapters analyse and translate. It does not survey oceanography for its own sake. It gives only what is needed to understand what the *detect* phase recovers and what the *translate* phase makes perceptible.

4.1 The Physical System

The Atlantic Meridional Overturning Circulation is the ocean’s conveyor of heat through the Atlantic. Warm, salty surface water flows north; in the cold subpolar seas it loses heat to the atmosphere, grows dense, and sinks; it then returns south at depth as a cold, deep current (Figure 4.1). This vertical overturning — surface flow one way, deep flow the other — is driven by differences in water density set by temperature and salinity, the process known as thermohaline circulation.

The consequence is climatic, not merely oceanographic. The northward surface flow carries roughly the heat of a million power stations into the North Atlantic, and much of the relative mildness of northwest Europe depends on it (Buckley & Marshall, 2016). Yet the system has no surface that an observer can see: it is a circulation in the ocean’s interior, spanning a basin, unfolding over decades. This is precisely the gap the thesis addresses — a process of great consequence and no everyday visibility.

4.2 Bistability and Tipping

What makes the AMOC a *tipping element*, rather than a system that simply weakens in proportion to warming, is a feedback first captured by Stommel (1961) in a simple two-box model of the ocean.¹ The mechanism is a loop: the overturning carries salt northward, salt keeps northern water dense, and dense water keeps sinking, sustaining the overturning. Add enough freshwater to the North Atlantic — from melting ice, for instance — and the loop can run in reverse: fresher water is less dense, sinking weakens, less salt is carried north, and the circulation slows itself further.

Because of this feedback, the AMOC has two stable states — vigorous overturning, and a much weaker one — for the same conditions. A gradual push can therefore trigger an abrupt switch, and reversing the push does not simply reverse the switch (Figure ??). This property, called hysteresis, is why the system’s behaviour is non-linear and why it resists intuition: cause and effect are not proportional. Communicating that non-linearity, which no static curve conveys, is the representational challenge at the centre of this thesis.

¹ Stommel represented the ocean as two boxes, equatorial and polar, exchanging water at a rate set by their density difference. The salt-advection feedback in his equations yields two stable equilibria for the same external forcing — the mathematical signature of bistability.

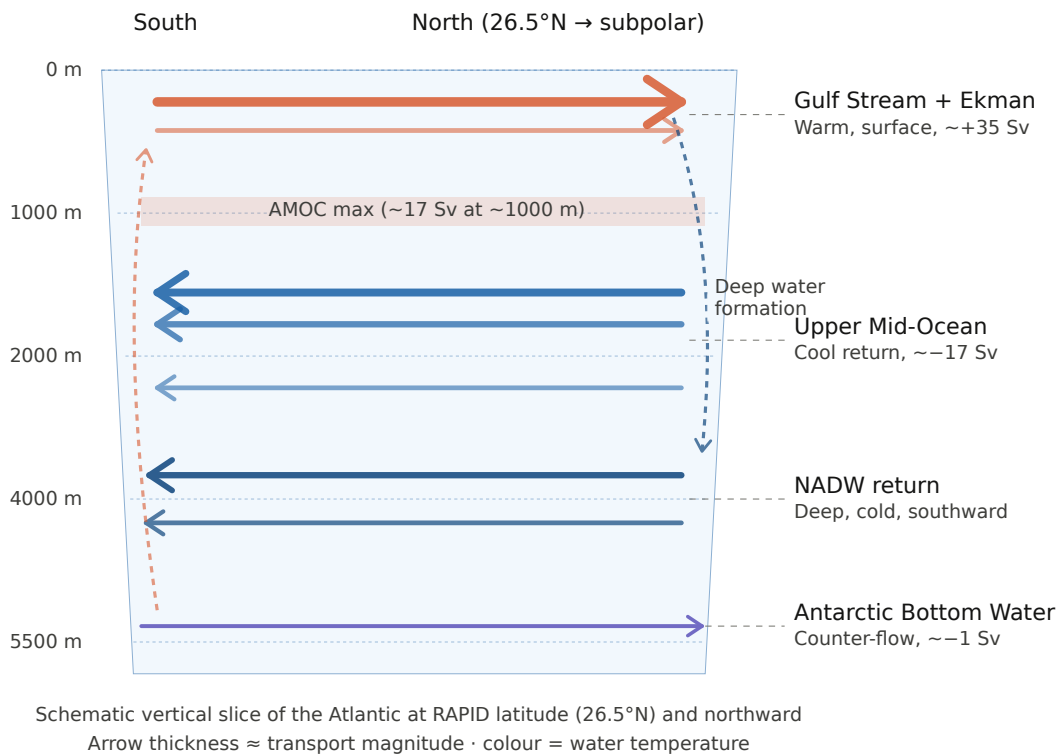


Figure 4.1: The Atlantic overturning circulation in cross-section. Warm surface water flows north (magenta), releases heat to the atmosphere in the subpolar seas, grows dense and sinks, and returns south as a cold deep current (teal). The vertical overturning is driven by density differences of temperature and salinity. The circulation has no visible surface expression — the invisibility the thesis sets out to address.

4.3 Measuring the Circulation

To analyse the AMOC, its strength must be expressed as a number. That number is the volume of water the overturning moves, measured in *sverdrups* ($1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$ — a million cubic metres per second).² A healthy overturning at mid-latitudes runs at roughly 15–20 Sv.

Crucially for what follows, the circulation is not a single number but a *profile*: transport varies with depth, strong near the surface where warm water flows north, reversing at depth where cold water returns south. This vertical profile, resolved over time, is the raw material the *detect* phase decomposes — the object from which machine learning recovers a compact latent structure in Chapter 6.

4.4 Observational Data

The AMOC is observed directly by the RAPID-MOCHA-WBTS array at 26.5° north, a line of moored instruments spanning the Atlantic that has measured the overturning continuously since 2004 (Ben I Moat et al., 2026; Frajka-Williams et al., 2023). RAPID is the empirical anchor of this thesis: a genuine, continuous record of a system that had previously been only inferred.

² Formally, the overturning streamfunction integrates the northward velocity across the basin and upward from the sea floor; its maximum over depth defines the AMOC transport at a given latitude.

Its one limitation is length. Two decades of data are short against a circulation that varies over decades to centuries, so the observed record cannot alone reveal the system's slow behaviour. To provide longer context, the analysis adds the Copernicus Marine GREP ensemble — a reanalysis that reconstructs the circulation before the observational period by combining models with assimilated data, at the cost of introducing model dependence and a spread between ensemble members. That spread is not merely a caveat: it becomes, in the *translate* phase, the raw form of uncertainty the representation must make perceptible. Together, observation and reanalysis are the two datasets the next chapter analyses.

Chapter 5

Methodology

Addressed: Approach · Research Design & Workflow · Hypotheses as Drivers · Data Collection & Analysis · Justification

This chapter explains the logic that holds the thesis together: what research approach was chosen, why this particular combination of methods, and how data becomes, step by step, an experiential representation that can be tested. It does not repeat the research questions and hypotheses set out in Chapter ??, nor the results reported later. It explains the reasoning that connects them.

5.1 Methodological Approach

The thesis asks whether a machine-learning-driven experiential representation helps non-specialists understand a complex climate system better than a conventional one. That question cannot be answered by analysis alone, nor by building something alone. It needs both: an artefact has to be built, and its effect has to be measured. This places the work within *Design Science Research* (DSR), the tradition made for research whose central contribution is a built artefact and whose knowledge comes from building and testing it (Hevner et al., 2004; Peffers et al., 2007). The artefact here is the experiential representation; the knowledge is what its making and testing reveal about turning an invisible system into something people can perceive.

DSR is a family of methods rather than one fixed recipe, and this thesis follows the widely used six-phase model of Peffers et al. (2007): identify the problem, define objectives, design and develop, demonstrate, evaluate, communicate. The thesis maps onto this model through its *detect–translate–evaluate* arc. *Detect* carries problem identification and the recovery of the material to be represented; *translate* carries the design and development of the artefact; *evaluate* carries its testing with participants; and the thesis itself is the communication. The arc is not a new methodology but a plain-language version of an established one, named for what the work actually does with a climate system.

Within this frame, quantitative and qualitative methods play distinct and complementary parts. The *detect* phase is quantitative: statistical and machine-learning analysis of ocean data. The *translate* phase is design-based: turning that analysis into perceptible form. The *evaluate* phase is a mixed-methods user study, mainly qualitative with descriptive quantitative support. These are not rival paradigms competing for the thesis; they are three instruments, each suited to a different phase of one design-science process. The stance that lets them sit together is *pragmatism* — judging knowledge by whether it works for a purpose rather than by whether it matches a single fixed reality.¹

One feature of DSR shapes the whole design and is worth stating plainly. Hevner’s rule is that an

¹ Pragmatism is the epistemology conventionally paired with design science and mixed methods (Peffers et al., 2007). It licenses treating a machine-learning latent structure as a valid substrate for representation not because it is uniquely “correct”, but because it is useful for making an invisible system perceptible; and it lets quantitative measurement and qualitative accounts count as evidence on the same practical question without forcing one into the other’s mould.

artefact must be evaluated, not merely built: usefulness that is claimed but not shown is not a contribution (Hevner et al., 2004). This is why the central research question is comparative — *to what extent, compared with* a conventional baseline — rather than a question a finished artefact would answer just by existing, and why the *evaluate* phase is essential rather than decorative.

5.2 Research Design and Workflow

The thesis runs as a pipeline in which the output of each phase is the input of the next (Table 5.1). It begins with the AMOC as a physical system (Chapter 4). The *detect* phase analyses two decades of ocean data to recover the system’s latent structure — a small set of coordinates that captures how it behaves. The *translate* phase takes those coordinates and, guided by documented perceptual principles, turns them into representations a non-specialist can experience. The *evaluate* phase puts those representations in front of participants and studies how they understand the system. Each arrow in that chain is a methodological commitment: the analysis constrains what can honestly be shown, and the representation is what the study then tests.

Table 5.1: The thesis pipeline. Each phase takes the output of the previous one as its input, so that the analysis constrains the design and the design is what the evaluation tests.

Phase	Method	Input	Output
Detect	PCA, autoencoder, anomaly & early-warning analysis	RAPID observations; Copernicus reanalysis	Interpretable latent structure of the system
Translate	Design grammar applied to recovered features; browser-based build	Latent structure from <i>detect</i>	Three representational conditions (static, dynamic, experiential)
Evaluate	Two-group study; think-aloud; reflexive thematic analysis	Representations from <i>translate</i>	Evidence on how non-specialists understand the system

The single thread running through the pipeline is *fidelity*. What the analysis recovers sets the limit of what the representation may claim, and what the representation shows is exactly what the evaluation measures. Nothing is added between phases that the previous phase did not justify. This is what makes the final artefact defensible as a translation of the science rather than an illustration inspired by it.

5.3 Hypotheses as Methodological Drivers

The hypotheses and research questions were formalised in Chapter ?? and are not restated here. What matters methodologically is how each kind of claim decided the method used to investigate it, because the three kinds call for different treatment.

The analytical hypotheses (H1–H3) are questions about what is in the data, so they are investigated by quantitative analysis whose results can confirm or overturn them: a decomposition to test how few modes capture the structure, a model comparison to test whether that structure is genuinely low-dimensional, and a corrected significance test to check whether an apparent warning signal survives honest scrutiny. The design propositions (DP2.1–DP2.2) are not true-or-false claims but design commitments, so they are

investigated by construction and defended against design criteria rather than tested for significance. The empirical expectations (E3.1–E3.2) concern how people understand, so they are investigated qualitatively, through what participants say and do, not through a statistical test the sample could not support. Matching each claim to the kind of evidence it can actually bear is itself a methodological decision, and it governs every choice in the sections that follow.

5.4 Data Collection and Analysis Methods

5.4.1 The analytical phase

The analysis draws on two sources of different standing. The RAPID-MOCHA-WBTS array at 26.5° north has directly *observed* the overturning since 2004 (Ben I Moat et al., 2026; Frajka-Williams et al., 2023); the Copernicus GREP ensemble is a *reanalysis* that reconstructs the circulation further back, at the cost of model dependence and ensemble spread (European Union-Copernicus Marine Service, 2019). The distinction governs their use: RAPID anchors the analysis in measurement, the reanalysis supplies longer context and the raw material for uncertainty. Every claim is framed to respect which of the two it rests on.²

Three methods then act on this data, each chosen for a specific reason; their results appear in Chapter 6. *Principal component analysis* reduces the high-dimensional vertical structure to a few coordinates, chosen because its components are linear and physically interpretable — and interpretability is a requirement here, since these coordinates become the substrate a non-specialist will inhabit, and one that cannot be explained cannot be defended. Because PCA is linear, a *non-linear autoencoder* is trained alongside it as a check: if the autoencoder reconstructs the data no better at equal dimensionality, the structure is genuinely low-dimensional and the interpretable coordinates lose nothing. This comparison is what lets the thesis justify, rather than assert, its choice of coordinates. Finally, *anomaly detection* and *early-warning analysis* characterise the record over time — the first to find the events a representation should convey, the second to test for the signals of an approaching transition, with the significance test corrected for the statistical dependence of overlapping windows so the phase cannot manufacture a signal from an artefact.

One boundary is set deliberately. A predictive layer — a model forecasting future transport — is excluded and left to future work, because the *detect* phase exists to recover structure for representation, not to predict, and a forecasting goal would shift the thesis away from the representational contribution it claims.

5.4.2 The design phase

The translation from structure to perception is not free invention. It applies the design grammar established in Chapter 3, which maps a system property to a perceptual channel through a documented cognitive principle. The method of this phase is to apply that general grammar to the specific features the analysis recovered, so that every design decision traces back to a principle and a source rather than to taste. Two commitments keep the translation honest. First, the three representational conditions carry the *same information* and differ only in *form*, so that any difference a participant shows can be attributed to form,

² Treating a reconstructed value as if it were observed would misrepresent the evidence, so the two are never conflated. This boundary is recorded in Appendix A.

not to being given more facts.³ Second, every value shown on screen derives from the analytical output, never from a scaling invented to look good — a rule that constrains the aesthetics to what the data support, recorded as a design boundary in Appendix A. The artefact is built for the browser so it runs on an ordinary laptop in a participant's own setting, which keeps the evaluation practical; its full construction is documented in Chapter 7.

5.4.3 The evaluation phase

The evaluation is an exploratory, two-group, between-subjects study with non-specialist participants. A control group sees only the conventional static condition (C1); an experimental group sees the dynamic and experiential conditions together (C2 and C3). Each person is in one group and sees one kind of representation, so the comparison is between two whole representational experiences rather than isolated features.⁴ Roughly 24 to 30 participants are split between the groups — realistic for the timeframe and sufficient for rich thematic analysis, but deliberately not sized for statistical inference.

The study is delivered as a self-guided online questionnaire, so that participants take part in their own setting and the representations reach an ordinary browser without the researcher present.⁵ It is structured as a short journey. It opens with brief background questions — age, occupation, and prior familiarity with the topic — which characterise the participant and let prior knowledge be accounted for in analysis. The participant is then taken through the sequences of their assigned condition, with comprehension questions placed between them, so that understanding is captured as it forms rather than only at the end. After the assessed part is complete, every participant is shown the fuller experiential sequence once, purely for interest; responses to it are not assessed and do not enter the comparison, so that participants who saw only the static version are not left without the experience.⁶

The comprehension questions are the study's evidence, and they lean on open-ended items, with closed items possibly added in support.⁷ The open responses are the primary evidence, analysed through reflexive thematic analysis (Braun & Clarke, 2006, 2019), which builds themes from what participants write rather than imposing them in advance. Any closed items are reported descriptively, not as a significance test: with this sample, directions and differences between the groups are honest, but p-values would not be. The full instrument is finalised close to data collection and reproduced in Appendix A.⁸

3 C1 is a conventional static representation reconstructed from how the AMOC is actually published; C2 adds interactivity and motion; C3 adds the experiential, machine-learning-driven layer. How they are grouped for the study is decided in the evaluation design below.

4 The experimental group's exposure differs from the control's in extent as well as kind, so the design does not isolate the effect of form from the effect of richer exposure. It compares two experiences as wholes — which is exactly what the central question asks. This residual confound is acknowledged, not hidden, and recorded in Appendix A; separating the conditions is future work.

5 Two parallel versions are prepared, in Italian and in English, so participants respond in their first language; open responses are analysed in the language of the answer to avoid translation distorting meaning. Two further versions split the groups, one showing the static condition and one the dynamic-experiential condition, giving four questionnaire variants in total.

6 This closing sequence functions as a debrief and a return to the participant, not as a data source. Framing it explicitly this way keeps it from acting as a confound: it falls outside the assessed journey by design.

7 Open questions reveal a participant's mental model of the system — how they describe its dynamics, its non-linearity, its uncertainty — rather than whether they can recognise a supplied answer; closed items, if used, give a signal that is directly comparable between the two groups. The exact mix is being finalised as part of the work still in progress.

8 No market analysis is conducted. The thesis contributes a representational method, not a product for a market, so the evidence that bears on it is how participants understand the system, not whether a market demands the artefact — a scoping decision, not an omission. Ethics is handled through informed consent, voluntary participation, anonymisation and GDPR-compliant storage; the topic is not sensitive, and the study proceeds only once the supervisor confirms the appropriate approval is in place. Full

5.5 Methodological Justification

The methods answer to one demand: making an invisible climate system perceptible without compromising its scientific accuracy. That demand pulls in two directions, and the design is built to hold both. On one side is quantitative rigour — the analysis must be honest about what the data show, which is why the early-warning test is corrected for dependence and why the choice of coordinates is validated against a non-linear model rather than assumed. On the other side is human perception — the representation must be legible to someone without training, which is why the coordinates must be interpretable and the design grounded in how people actually see and reason.

The link between the two sides is the thread of fidelity that runs through the whole pipeline. Interpretable coordinates connect the rigour of the analysis to the legibility of the design: they are faithful to the data and explainable to a person. The grammar connects design to cognition: each perceptual choice rests on evidence, not preference. And the comparative, qualitative evaluation connects the artefact back to its purpose: it asks not whether the representation is admired but whether it is understood. Each method was chosen because it carries one of these connections, and together they let the thesis do the difficult thing it set out to do — translate a system no one can see into something a person can grasp, without letting the translation drift from the science it stands on.

Chapter 6

The Machine Learning Substrate

*Addressed: Latent Structure · Manifold Geometry · Early Warning Signals · Anomalies
· Observation vs Reanalysis*

This chapter represents the core of the *detect* phase: the analysis that recovers, from the AMOC observational record, the latent structure the *translate* phase will later make perceptible. It delivers the objectives set out in Chapter ?? (O1.1–O1.3) and tests the analytical hypotheses H1–H3. As already stated, the aim is not a new claim about the ocean. A reader looking for an oceanographic discovery will not find one here, and that is by design. The aim is narrower and, for this thesis, more important: to establish with evidence that the AMOC’s vertical structure can be reduced to a small, faithful, interpretable set of coordinates. Everything the representation does in Chapter 7 rests on that substrate being trustworthy. The analysis therefore focuses on the data itself, rather than on the physical system it describes, and the results are framed in terms of their implications for the representation rather than their oceanographic significance.

The material is the RAPID-MOCHA-WBTS record at 26.5° north: the overturning streamfunction sampled at 307 depth levels, every day from April 2004 to March 2024. Three days near the end of the record are incomplete and are dropped, which leaves 14,596 daily profiles, each a vector of 307 numbers describing how the transport is distributed with depth.¹ Before anything else, each profile is centred on the record mean, so the analysis describes how the vertical structure *departs* from its average shape rather than simply recovering that average shape again.

6.1 Latent Structure & PCA Decomposition

The first objective (O1.1) is to find the dominant modes of the AMOC’s vertical variability using principal component analysis. The result is clear-cut: the variability is almost one-dimensional. A single mode accounts for 89.82 % of the total variance in the vertical structure, and two modes together account for 97.86 % (Table 6.1). From the third mode on, each mode adds less than 1.6 %, and the scree curve bends sharply between the second and third modes — the familiar elbow that marks where structure ends and noise begins. This confirms H1, and by a wide margin: the methodology chapter expected two or three modes above 80 %, and the data give two modes above 97 %.

Both modes have a clear physical meaning and this is important because they provide a tangible interpretation of the abstract components. The first mode, PC1, has the same sign at every depth and peaks near 1150 m, exactly where the average profile peaks. When its value rises, the whole overturning cell grows stronger while keeping its shape; when it falls, the cell weakens as a whole. PC1 is the *intensity* of the circulation, and it is close to ninety per cent of what the AMOC does from one day to the next. The

¹ The depth grid is close to uniform (spacing 19.33–19.87 m, median 19.59 m; a max/min ratio of 1.03), so no layer-thickness weighting is applied: with levels this evenly spaced, weighting would scale every level by an almost identical constant, which PCA absorbs without any effect on the loadings or the variance shares.

Table 6.1: Variance explained by each principal component of the AMOC vertical profiles. Two modes reach 97.86 %; the tail beyond PC2 is noise.

Mode	Variance	Cumulative
PC1	89.82%	89.82%
PC2	8.03%	97.86%
PC3	1.57%	99.42%
PC4	0.37%	99.80%
PC5	0.12%	99.91%
PC6	0.05%	99.97%

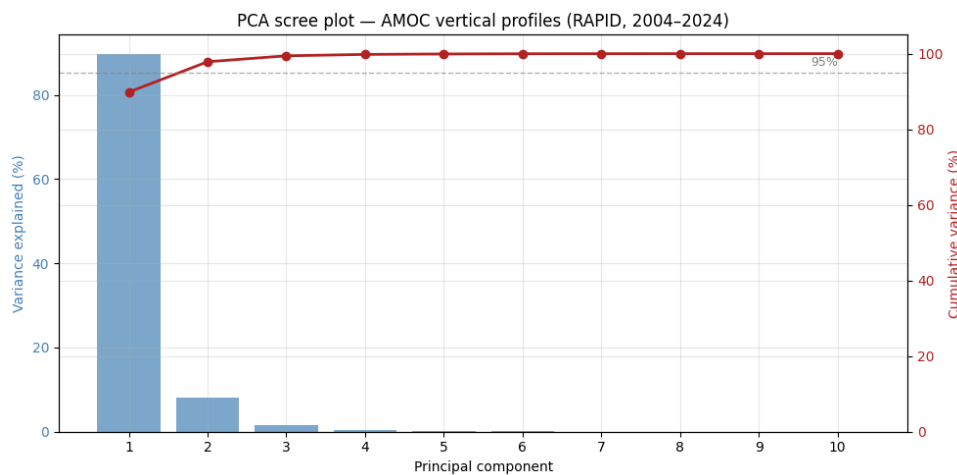


Figure 6.1: The vertical variability of the AMOC is almost one-dimensional. Variance explained by each principal component of the 307-level vertical profiles. The first mode alone explains 89.82 % and the first two together 97.86 %; the sharp elbow after the second mode marks where structure gives way to noise. This is the two-coordinate latent space the experiential representation is built on.

second mode, PC2, changes sign once as it goes down through the water column: it strengthens the cell at one depth while weakening it at another. It is the *vertical redistribution* of the circulation — a smaller but genuine shift in the shape of the overturning around its mean.

This is where the analysis first touches the thesis directly. It would be easy to assign meanings to these modes after the fact, but that is not what happens here. The intensity mode is not a story imposed on the data; it is recovered from the data and then confirmed against a known event, which is a stronger position to argue from. Followed through time, PC1 drops sharply during 2009–2010 — the most thoroughly documented weakening in the observational record — reaching a period mean PC1 score of -23.97 against a whole-record mean of zero.² That a single line can still carry an event first seen in the full depth–time field is a small but real proof of the thesis claim — that this system can be re-represented in fewer dimensions without losing what matters. The PC1–PC2 plane, holding 97.86 % of the vertical anatomy in just two coordinates, is the latent space along which the profiles will breathe faithfully through time in the experiential condition of Chapter 7.

² The same 2009–2010 event shows up independently in the raw depth–time field, in this intensity series, and later in the anomaly detector: three methods, one physical episode.

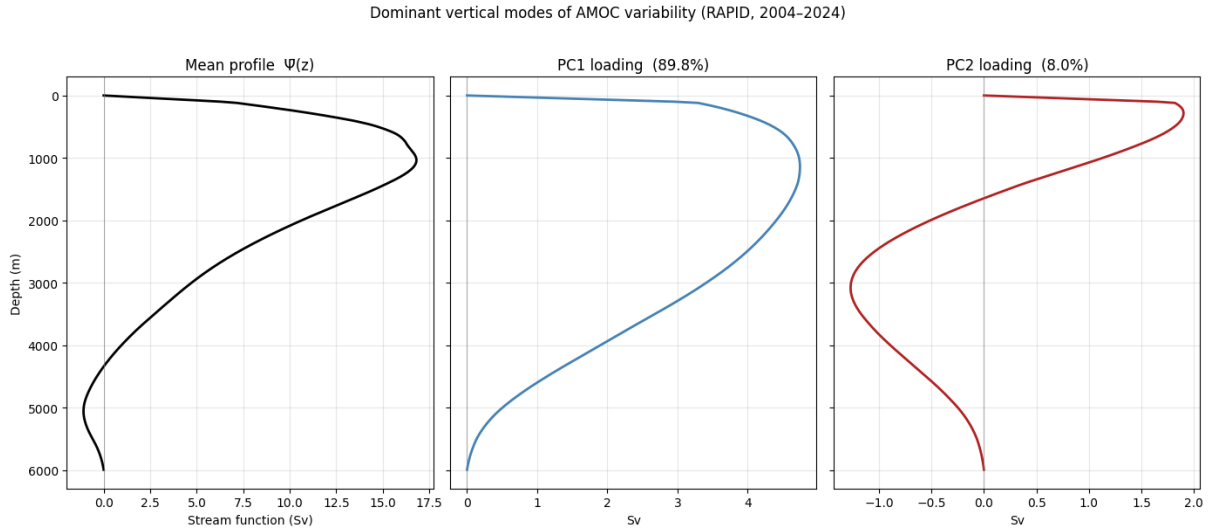


Figure 6.2: The two modes that explain the AMOC’s vertical anatomy. Each curve is a loading profile plotted against depth. PC1 (intensity) is a *monopole*: the same sign at every depth, peaking near 1150 m where the mean profile peaks, so a change in its score strengthens or weakens the whole overturning cell coherently. PC2 (redistribution) is a *dipole*: it changes sign once with depth, strengthening the cell at one level while weakening it at another. Loadings are Sv-scaled for display. Together these two shapes carry 97.86 % of the vertical variability.

6.2 An Essentially Flat Manifold Geometry

A latent space is only useful if it provides a faithful representation of the underlying system. Since PCA is a linear method, an important question is whether a simple linear representation is sufficient to capture the structure of the AMOC data. If the underlying geometry were strongly non-linear, a linear projection could oversimplify it and lose important information. The second objective (O1.2) therefore tests this assumption by comparing PCA with a non-linear autoencoder trained on the same 14,596 AMOC profiles. Both models are evaluated using the same number of latent dimensions. If the latent structure were genuinely non-linear, the autoencoder would be expected to reconstruct the data more accurately. If not, PCA should perform equally well.

The results clearly support the latter interpretation. Across latent dimensions from one to five, the non-linear autoencoder never outperformed PCA. Instead, PCA consistently achieved slightly better reconstruction accuracy, with differences ranging from -0.06 to -0.15 percentage points in its favour (Table 6.2). At the two-dimensional representation used throughout this thesis, PCA reconstructed 97.86 % of the variance, compared with 97.77 % for the autoencoder, a difference of less than one tenth of a percentage point.³ These findings confirm H2. Even when given the flexibility to learn complex non-linear transformations, the autoencoder found no meaningful structure beyond that already captured by the linear PCA representation.

This result is important because it justifies the use of the latent coordinates throughout the *Translate* phase of the thesis. If the underlying manifold had been strongly curved, representing the system using the first two principal components would have introduced distortions by forcing a non-linear structure into a linear space. Instead, the results show that the latent manifold is effectively linear. As a consequence, the first two principal components provide a representation that is both highly interpretable and highly

³ The surface and seafloor levels were excluded from the autoencoder input because the streamfunction is fixed to zero at both boundaries by mass conservation and therefore contains no learnable variance.

Table 6.2: Reconstruction of the vertical profiles by a non-linear autoencoder against linear PCA, at matched latent dimension k . The gap is negative at every k : PCA is never beaten, so the manifold is flat.

k	PCA cum. %	AE R_{Sv}^2 %	Gap (pp)
1	89.82	89.67	−0.15
2	97.86	97.77	−0.08
3	99.42	99.36	−0.06
4	99.80	99.74	−0.06
5	99.91	99.83	−0.08

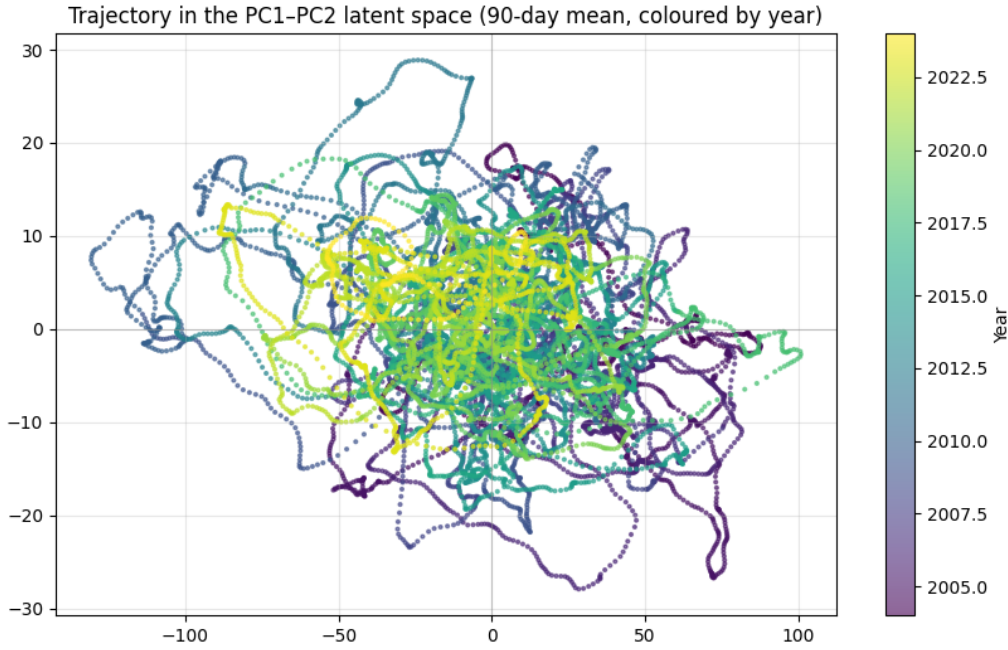


Figure 6.3: The manifold of AMOC vertical profiles is flat. Each point is one day, placed by its two latent coordinates — PC1 (intensity) and PC2 (vertical redistribution). A non-linear autoencoder trained on the same profiles reconstructs them no better than linear PCA at any latent dimension (gap -0.06 to -0.15 percentage points; see Table 6.2), confirming that two straight coordinates capture the structure without distortion.

faithful to the original data. The coordinates can therefore be meaningfully interpreted as distinct modes of AMOC variability without sacrificing representational accuracy. In this case, interpretability comes at virtually no cost in fidelity, making the proposed representation both scientifically defensible and suitable as the basis for the experiential visualisations developed in the following chapter.

6.3 Early Warning Signals & Null Result

The third objective (O1.3) investigates whether the record contains early warning signs of an approaching tipping point. According to the theory of critical slowing down, a system approaching a bifurcation is expected to show increasing variance and increasing lag-1 autocorrelation (Dakos et al., 2012; Scheffer et al., 2009). Testing this hypothesis requires particular care, and this is where the analysis highlights an important methodological issue.

At first glance, the results appear very strong. The rolling variance of the intensity series shows an increasing trend, with Kendall's $\tau = +0.078$ and a p-value of 2×10^{-42} ; similarly, the rolling autocorrelation

increases with $\tau = +0.088$ and a p-value of 1×10^{-52} . Taken directly, these values would suggest an extremely strong statistical signal. However, this interpretation would be misleading.⁴

The reason is that the rolling windows overlap substantially, so the thousands of calculated values do not represent independent observations. This is a common statistical pitfall: the software returns a very small p-value without accounting for the dependence between the input points, making the result appear more certain than it actually is. When the number of independent observations is considered instead — approximately 19.5 non-overlapping 730-day windows — and the statistical test is corrected accordingly, the apparent trends disappear. The corrected p-value for the variance trend is **0.634**, while for the autocorrelation trend it is **0.595**. Neither result is statistically significant (Table 6.3). The initial strong signal was therefore caused by treating dependent observations as if they were independent.

Table 6.3: Kendall’s τ trend tests for the two classical early-warning signals. The naive p-values treat overlapping windows as independent; the corrected p-values count the ~ 19.5 independent windows the record contains. Neither trend is significant.

Signal	Kendall τ	Naive p	Corrected p	Verdict
Rolling variance	+0.078	2×10^{-42}	0.634	not significant
Rolling AR(1)	+0.088	1×10^{-52}	0.595	not significant



Figure 6.4: No rigorous early-warning trend. Rolling variance and lag-1 autocorrelation of the intensity series over the record. Both oscillate without a sustained climb. Their naive trend p-values ($\sim 10^{-42}$) are artefacts of heavily overlapping windows; corrected for the ~ 19.5 independent windows the record actually contains, the trends are not significant ($p = 0.634$ and 0.595 ; Table 6.3).

The final conclusion is therefore a *null result*. Once the dependence structure of the data is properly considered, the classical early-warning indicators cannot be distinguished from noise. This supports H3 in its null form and represents an important methodological contribution rather than a limitation. A positive

⁴ Kendall’s test assumes that observations are independent. In this analysis, rolling statistics are calculated using heavily overlapping 730-day windows, meaning that consecutive values are highly dependent. This effect is amplified by the strong day-to-day autocorrelation of the original series (0.943).

result obtained by incorrectly interpreting the statistics would have been more appealing, but it would not have been reliable. Demonstrating that an apparently strong signal disappears when statistical dependence is properly addressed reflects the methodological care at the core of this thesis.

6.4 Anomaly Detection

Early-warning analysis looks for slow *trends*. A different question is about individual *events*: which moments depart from the normal behaviour of the system. An Isolation Forest applied to the daily PC1–PC2 coordinates identifies 5 % of the record, corresponding to 730 days, as anomalous. Looking at when these days occur reveals an important pattern: they are concentrated in 2009–2010 (158 days), 2012–2013 (100 days), and 2018 (66 days). However, the anomaly score does not increase over time. The unusual moments are distributed across the record rather than accumulating toward the end, providing another indication that the system is not showing a progressive movement toward a threshold.

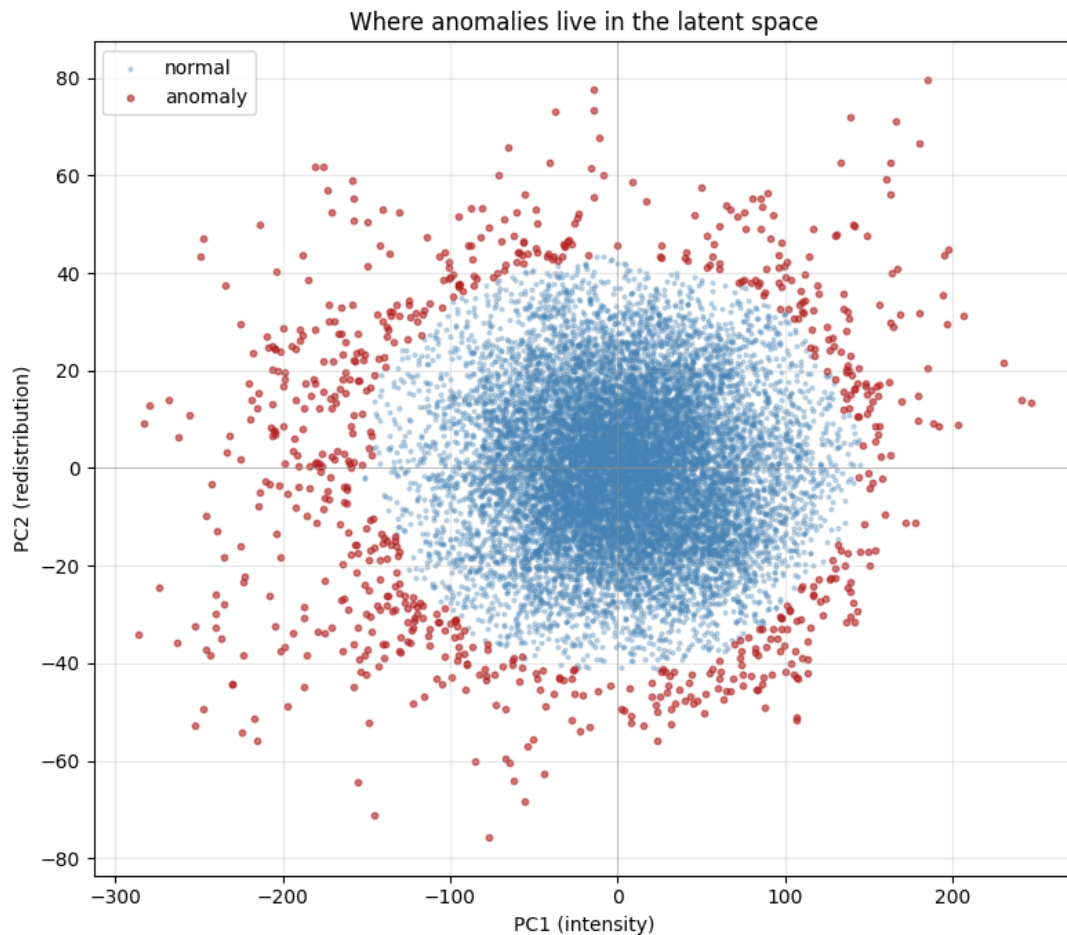


Figure 6.5: Discrete anomalous events, spread across the record. The Isolation Forest flags 5 % of days (730) as anomalous in the PC1–PC2 plane. They cluster in 2009–2010 (158 days), 2012–2013 (100) and 2018 (66), with no trend over time — corroborating the early-warning null from a different method. These moments become perceptible texture (grammar principle P3) in Chapter 7.

Two aspects are important for this thesis. First, this analysis supports the null result from a completely independent perspective: an algorithm that does not rely on early-warning theory also finds no gradual shift, but rather isolated events distributed across the twenty-year record. It reaches the same conclusion

through a different method.⁵ Second, these anomalies provide the basis for a specific design decision. Following the grammar introduced in Chapter 3, departures from the normal regime can be translated into local changes in perceptual texture (principle P3). The moments identified by the detector are therefore the same moments that the experiential representation will transform into perceptible changes — meaning that anomaly detection is not only a result about the AMOC, but also a direct input for how the AMOC is represented.

6.5 Observation Against Reanalysis

The analysis so far has relied on RAPID observations. The final step compares them with the Copernicus GREP reanalysis ensemble over the 237 months they share. This comparison serves two purposes: characterising how well the reanalysis reproduces the observed signal, and identifying the uncertainty that the *translate* phase needs to communicate. The two datasets show moderate agreement. Their correlation is 0.754, with a small bias of +0.584 Sv (RAPID being slightly stronger) and a root-mean-square difference of 2.300 Sv. Both remain close to 17 Sv and show similar variations over time. In other words, the reanalysis captures the overall behaviour of the circulation, even when it does not reproduce the exact value of each individual month.

The most informative result concerns uncertainty coverage. RAPID falls within the reanalysis ensemble's own ± 1 standard-deviation range in only 40.1 % of months.⁶ A model ensemble with a realistic uncertainty range should capture the observation more frequently. This low coverage indicates that the ensemble is *overconfident*: its reported uncertainty is smaller than the real error. This point is important for the thesis because the ensemble spread is exactly what the *translate* phase converts into perceptible uncertainty, following design-grammar principle P4. Understanding that this uncertainty is underestimated is therefore part of representing it honestly: the visualisation can communicate the model's uncertainty while making clear that the real uncertainty is likely larger.

6.6 Analytical Summary

The analytical phase provides the *translate* phase with a documented substrate rather than an assumed one. The vertical structure of the AMOC can be represented through two interpretable coordinates that explain 97.86 % of its variability: intensity and vertical redistribution. These coordinates are not only a convenient simplification but a faithful representation of the system: a non-linear model confirms that the underlying manifold is effectively flat, meaning that the transformation does not introduce relevant distortion. The record does not show robust early-warning trends, but it does contain clear discrete anomalies that can become perceptual texture. The comparison with the reanalysis also provides an uncertainty estimate that is physically meaningful, although too narrow, and therefore ready to be translated into perception. These findings are not only descriptions of the ocean system. Each one becomes a justified input for the next phase: a compact, faithful, interpretable, event-aware, and uncertainty-carrying substrate that allows the *translate* phase to make an invisible system perceptible without misrepresenting it.

⁵ The anomaly clusters correspond to independently identified features — the 2009–2010 weakening and the variance peaks of 2012–2013 and 2018 — with 41.4 % of anomalous days occurring in the highest quarter of local variance. Different methods reveal the same physical behaviour.

⁶ If the ensemble spread represented a well-calibrated estimate of its own uncertainty, the observation should fall within the one-standard-deviation range much more often. A lower coverage indicates that the spread is too narrow compared with the actual errors.

Part II

TRANSLATE

From Latent Structure to Experiential Representation

Chapter 7

Designing Experiential Representations

Addressed: Prototype · MVP · Visualizations

Chapter 6 established what there is to represent. This chapter establishes how it is represented, and why those choices are defensible. It documents the three artefacts of the *Living Atlantic* project, the framework governing all three, and the design grammar offered as this thesis’s transferable contribution under objective O2.2.

7.1 From Latent Structure to Perceptual Form

Objective O1.2 compared a non-linear autoencoder against linear PCA at matched latent dimensionality. Two linear modes account for 97.86% of the variance in the AMOC vertical profiles, the first alone for 89.82%, and the autoencoder never exceeds the linear decomposition at any dimensionality from one to five, tracking it within 0.06 to 0.15 percentage points. The manifold on which the observational record lives is essentially flat.

Had the manifold been curved, moving through the plane recovered by PCA would have passed through states the ocean never occupied. Because it is flat, the coordinates are an accurate description of how the ocean truly behaves: each month is a point, and animating along the path between months shows real states rather than plausible interpolations.¹ Hypothesis H2 is therefore not a negative result but the foundation of the representational phase. Without it, the experiential layer would have no epistemic ground.

This is also the sense in which machine learning operates representationally rather than predictively here. The models of Chapter 6 do not forecast; they propose a coordinate system. Following Iten et al. (2020), extended to interactive exploration by Camps-Valls et al. (2021), the latent representation is treated as an object a human interpreter may examine and inhabit. What this chapter adds is making the latent space perceptible to a viewer who does not know it exists (Sacha et al., 2017b).

7.2 The Translate Framework

Five commitments govern all three scenes. They are stated once here and applied below.

7.2.1 Informational Equivalence by Shared Provenance

Objective O2.1 requires that the two conditions — conventional static (C1) and dynamic-experiential (C2+C3) — encode the same data, so that any difference observed in evaluation is attributable to form rather than content. B. Tversky et al. (2002) showed why this is not a formality: reviewing the experimental literature on animation, they found the apparent superiority of animated graphics dissolving wherever the

¹ Under the Baldi–Hornik result, any autoencoder gain over PCA is attributable to curvature alone; the near-zero gap confirms the manifold is flat enough for linear coordinates to stand in for it faithfully (Baldi & Hornik, 1989).

animated condition carried information the static one did not. Their equivalence principle forbids the comfortable assumption that the experiential condition will win².

The final study adopts a between-subjects design: each participant is assigned to a single condition and experiences all three scenes within that condition.³ This removes the carry-over effects that would arise if participants experienced both conditions in sequence, although it also means that direct within-person comparisons are no longer possible. For the exploratory qualitative design adopted in this thesis, involving a limited number of participants and analysed through think-aloud and reflexive thematic analysis (Braun & Clarke, 2006), this trade-off is appropriate. The aim is not to estimate causal effects or effect sizes, but to compare how participants interpret the two representational approaches under equivalent informational conditions.

Informational equivalence is therefore no longer guaranteed by the file structure itself, but by a shared provenance pipeline. Both conditions are generated from the same data bundle, reconstruct every AMOC profile using the same reconstruction function, derive shared visual elements from the same routines, and document every perceivable element in a common provenance ledger.⁴ This approach makes informational equivalence transparent and verifiable.

7.2.2 Fidelity

Every value on screen derives from the bundle. Nothing is rescaled for visual effect, and where an aesthetic preference conflicted with the data, the data won.⁵ The reconstruction is one equation, shared by every scene and both conditions:

$$\Psi(z, t) = \bar{\Psi}(z) + pc_1(t) \cdot B_1(z) + pc_2(t) \cdot B_2(z) \quad (7.1)$$

where $\bar{\Psi}$ is the record mean, B_1 and B_2 are the unit basis vectors of the decomposition and pc_1 , pc_2 the latent coordinates of month t .⁶

The equivalence claim can be checked against this equation directly. The static condition obtains the empirical mean by evaluating the shared reconstruction with both coefficients set to zero, which returns the stored mean profile exactly, with a maximum absolute difference of zero rather than merely negligible. C1 therefore follows the same computational path as the experimental condition while displaying only observed data. The model enters when the curve begins to move.

² An earlier version of the project enforced this equivalence through a shared document architecture. Although the study was later redesigned as a between-subjects experiment, the same principle remains: both conditions are generated from the same data bundle and reconstruction pipeline, differing only in the representational choices applied

³ Administered files are `column-static.html`, `theengine-static.html` and `thethreshold-static.html` (C1), and their `-experiential` counterparts (C2+C3). A third file per scene exposes all conditions behind a toggle for demonstration and is not administered.

⁴ `amoc_translate_bundle.json` (schema v1.0.0), `shared/reconstruct.js` and `shared/bundle.js` respectively; the ledger is Table C.1 in Appendix C, which also documents the module architecture and the repository layout.

⁵ The places where this cost something are recorded in Appendix ??.

⁶ The bundle also carries EOF loadings — the same modes scaled into Sverdrups for display of mode shape only. They must never be passed to Equation 7.1; see Appendix C.

7.2.3 Machine Learning as a Substrate

The machine-learning contribution is structural, not generative. The model does not draw the profile curve, colour the particles or produce the mean state; the ledger records those as classical.⁷ What the model supplies is the latent coordinate system along which the profile moves through time, and the identification of the moments that depart from it. The originality claimed here is therefore not a new algorithm but an empirical bridge: a learned latent space used as the substrate on which a complex system is made perceptible to a non-expert, with the model's role confined to where the data licenses it and declared element by element.

7.2.4 The Design Grammar

Objective O2.2 asks for a design grammar mapping machine-learning-derived features onto perceptual dimensions, grounded in the pre-attentive processing and embodied cognition of §3.2 and reusable beyond this case. The grammar is stated in Table 3.1 and is not restated here.⁸ Each scene records which principles it realises, so that a mapping which fails in evaluation falls on that mapping — and through it on the principle — rather than on the artefact as a whole.

7.2.5 Symmetry of Framing

Within each scene, both conditions import the same framing modules byte-identically: an introduction carrying the scene's driving question, and a context panel giving the data source and a plain-language account of what is measured.⁹ The rule governing this material is that both groups may be taught to read the instrument, but neither may be told the reading. Participants are told what Ψ measures and what a Sverdrup is, because a viewer who cannot read the units cannot articulate anything. They are never told what the structure means — not that the limbs oppose each other, not where the boundaries lie. If the viewer must be told, the translation has failed, and the evaluation would measure the framing text rather than the representation.

Each scene's driving question is the instrument that makes this measurable. It is posed identically to both groups, answered by neither document, and kept visible throughout exploration, so that participants know what to look for without being told what they will find.¹⁰ It is what expectation E3.1 tests: the same words for every participant, with only the representational form differing in what it affords by way of an answer. The symmetry is not a courtesy to the control group; it is what makes the comparison mean anything at all.

⁷ The particle field is a derivative of Ψ (Equation ??) that any analyst could reproduce with a difference operator. The model touches only the coordinate system and the anomaly scores.

⁸ Its six principles, in brief: P0, most important information to the most accurate channel (Bertin et al., 1983; Sacha et al., 2017a); P1, ordered magnitude to spatial position (Cleveland & McGill, 1984); P2, dynamics to real motion (Cronin et al., 2009; Sterman, 2008); P3, anomaly to density and texture (Bertin et al., 1983; Ware, 2021); P4, uncertainty to animated variability; P5, secondary ordered variable to sound (*The Sonification Handbook* | Edited by Hermann, Hunt, Neuhoff, n.d.).

⁹ `shared/intro-*.js` and `shared/context-*.js`, one pair per scene. Identity is by import rather than duplication, so the two documents cannot drift apart through an edit to one of them.

¹⁰ The three questions are reproduced in Appendix B in the form administered. Keeping the question visible rather than showing it once is a design decision: without a stated task, a moving representation reads as decoration rather than enquiry.

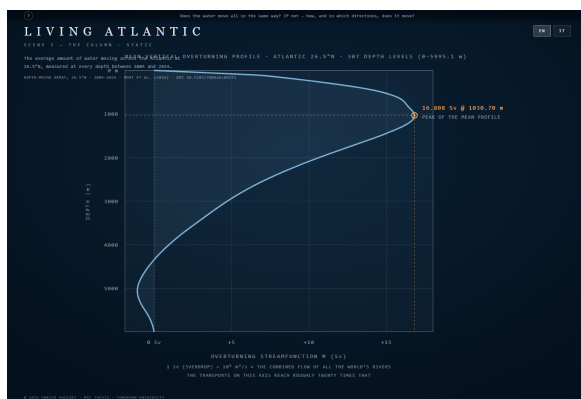
7.3 Three Scenes, One Descent

The three scenes are made to be experienced in order, and the order is interpretive rather than aesthetic. It follows the vertical image schema that Lakoff (1999) identify as a pre-conceptual cognitive resource — downward as deeper, more hidden, more fundamental. A viewer who meets the latent plane of Scene II without first inhabiting the column of Scene I has no referent for the coordinates; one who meets the ensemble spread of Scene III without having watched the observed trajectory has no baseline against which disagreement registers. Each scene is the condition of the next one’s intelligibility.

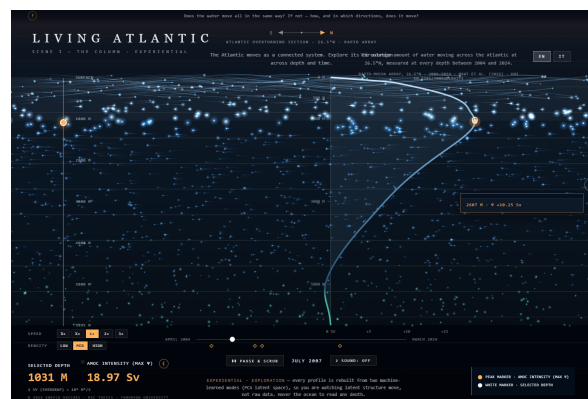
The scenes are nonetheless separate documents, so that each think-aloud session presents one scene in one condition with no accumulated interaction state.¹¹ Each is described below by the same six criteria: goal, representational strategy, mapping, conditions, fidelity and expected contribution.

7.3.1 Scene I: The Column

Driving Question. *Does the water move all in the same way? If not — how, and in which directions, does it move?*



(a) Control (C1): the canonical static profile.



(b) Experiential (C2+C3): the same column set in motion.

Figure 7.1: The two administered conditions of Scene I, generated from the same data bundle through the same reconstruction function. The experiential condition introduces temporal behaviour while preserving the underlying structure of the static representation.

Goal. To establish that the Atlantic at 26.5°N is not a current with a location but a quantity distributed over depth, whose parts move in opposite directions at the same moment. A viewer who leaves knowing only that the circulation is strong or weak has not received the claim. The driving question is whether the water moves all in the same way and, if not, how and in which directions.

Representational strategy. The canonical profile contains the counter-flow but does not display it. To see opposing motion in the curve, a reader must know that flow direction at any depth is given by the *slope* of the line rather than its height — where the curve rises water moves north, where it falls, south — and must read that slope by eye (Figure ??). Specialists do this without noticing they are doing it;

¹¹ Continuity is carried by the written framing preceding each scene and by the order of administration, not by animated transitions between them.

the literature on non-expert graph reading finds the gap to be the rule (Harold et al., 2016; Lang, 2014). Testing 110 decision-makers from 54 countries, Fischer et al. (2020) found a counter-intuitive IPCC graph systematically misread, with high confidence in the wrong reading. Scene I therefore does not help the viewer compute the derivative; it makes the derivative unnecessary, replacing the inference with motion.

Mapping. P0, namely the main information in the scene, is the vertical structure. P1 assigns the profile to spatial position across 307 depth levels spanning 0–5995.1 m.¹² P2 assigns the counter-flow to a particle field advected by the local meridional velocity $v \propto \partial\Psi/\partial z$, and the passage of time to the morphing of the curve along the latent trajectory. P3 assigns anomalous months to texture disturbance, scored by the IsolationForest of Chapter ???. P5 assigns pc_1 to a synthesised pulse, off by default. P4 is absent: the scene shows one trajectory and has no uncertainty to encode.

The particle field is driven by the derivative and never by Ψ itself. Because velocity is the vertical derivative of the streamfunction, it vanishes wherever Ψ is stationary — in the mean profile the peak of 16.808 Sv sits at 1030.70 m, and the first level at which the mean velocity has turned southward is the level immediately beneath it, at 1050.4 m.¹³ Driving the display from Ψ would move parcels fastest exactly where the true velocity is zero, and could not depict an overturning at all. The two boundaries do not coincide and must not: $\Psi \rightarrow 0$ marks where cumulative transport vanishes, at 4346.1 m; $v \rightarrow 0$ marks where local flow reverses, at 1050.4 m and 5065.4 m. Their separation of roughly 719 m (Figure ??) confirms the distinction is physical rather than pedantic.

Conditions. C1 is the canonical idiom: Ψ against depth, depth downward, publication grid, peak marked, and nothing else — no decomposition, no variance figures, no model output. C2+C3 is the same column set in motion (Figure ??). C1 is not a weakened comparison: it is drawn from the same bundle through the same reconstruction function at the same 307 levels, and its peak marker reports 16.808 Sv at 1030.70 m, against the 16.9 Sv at 1030 m published for this array (Smeed et al., 2018).

Fidelity. Every profile in either condition is rebuilt through Equation 7.1, and the identity that makes C1 empirical is the one recorded in §7.2.2.

Where the model enters. Scene I is the most physical of the three and the one where the model is least visible. The contribution is twofold. The model supplies the coordinates along which the profile can move through time: the static figure is an average over twenty years and averages do not move, whereas pc_1 and pc_2 make each month a point, so that walking between points reconstructs each month in turn. The motion is not an animation added to a figure but the model traversing its learned trajectory, and H2 is what makes that traversal faithful rather than convenient. The model also identifies which months depart from the record, and those become perceptible as texture rather than as a footnote. Everything else is classical: the particle field is a derivative, the axes come from the depth grid, the peak readout is the maximum of an array.

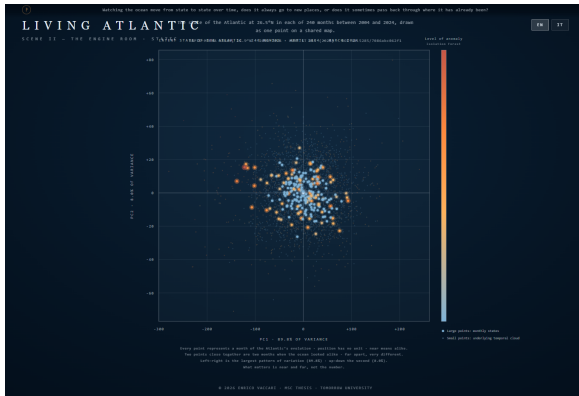
¹² The levels are almost uniformly spaced, at a mean interval of 19.59 m and a maximum-to-minimum ratio of 1.03. The sampling is essentially uniform rather than denser near the surface, and the display corrects for no non-uniformity that is not in the data.

¹³ Velocity anchors are reported at grid resolution: the derivation returns the first level where v changes sign rather than an interpolated zero. At 19.59 m spacing this falls below the resolution of the product. The Ψ zero-crossing is reported both ways: 4346.1 m at grid resolution, 4340.85 m interpolated.

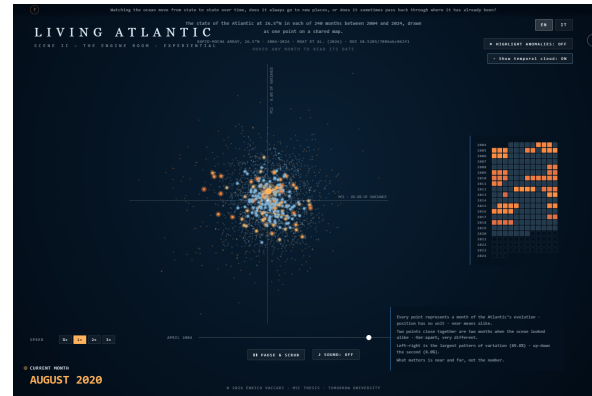
Expected contribution. C1 shows where the Atlantic is strong on average. C2 shows what C1 contains but cannot display. C3 adds the time the model learned: the column swells toward 24.26 Sv in January 2018 and collapses toward 5.71 Sv in March 2013, some 3.5σ below the mean of the monthly maxima, and the peak migrates as it goes — 1070.2 m in the strongest month, 991.1 m in the weakest (Figure ??).¹⁴ The claim is not that the experiential condition is more attractive, but that the static idiom is systematically misread, and that what it fails to convey — structure that must be inferred, change that must be imagined — is what motion and latent time deliver directly.

7.3.2 Scene II: The Engine Room

Driving Question. *Watching the ocean move from state to state over time, does it always go somewhere new, or does it sometimes pass back through where it has already been?*



(a) Control (C1): the static latent-space representation.



(b) Experiential (C2+C3): the evolving trajectory through latent space.

Figure 7.2: The two administered conditions of Scene II, generated from the same data bundle through the same reconstruction function. The static condition presents the latent structure as a fixed configuration, while the experiential condition reveals its temporal evolution.

Goal. To make the latent plane of Chapter ?? an inhabitable space rather than an analyst’s tool. Each month of the record is a point, and proximity means similarity. The claim the scene must deliver is that the system, left to wander, returns: it revisits states it has occupied before.¹⁵ The driving question asks whether the ocean, moving from state to state, always goes somewhere new or sometimes passes back through where it has already been.

Representational strategy. The canonical way to show a latent space is a scatter of points coloured by time, the figure any dimensionality-reduction paper publishes (Sacha et al., 2017b). That figure contains recurrence but cannot show it: two points may sit almost on top of each other while lying years apart, and a static cloud gives the eye no way to know that the path left one and later arrived at the other. Recurrence is

¹⁴ The migration is the second mode at work: pc_1 sets how much water the column carries, pc_2 redistributes it vertically. An earlier draft of this thesis asserted that the depth of maximum transport is stable while its magnitude varies; evaluated across the record, that is not the case.

¹⁵ This recurrence characterises the dynamics of the present observational record, in which the AMOC oscillates within a stable regime shaped by seasonal forcing and internal variability. It is not evidence that the system cannot undergo long-term transitions.

a property of the trajectory, not of the point set. Scene II therefore walks the path in time and accumulates the visited states as a cloud, so that a viewer sees the current month pass close to where it has been.

Mapping. P1 assigns each month to position in the $pc_1 \times pc_2$ plane, the axes labelled with the variance each mode explains. P2 assigns time to real motion along the path, and the magnitude of month-to-month change to the visible violence of each step: the largest steps are four times the median, and that variation is signal rather than noise to be smoothed.¹⁶ P3 assigns anomalous months to agitation of the marker, scored by the same IsolationForest. P5 assigns pc_1 to sound. P4 is again absent.

Conditions. C1 is the canonical scatter: the 240 months in the plane, coloured by year through an ordered ramp, static and unannotated. C2+C3 is the same points, the same data and the same colours, set in motion. Behind the monthly points, both conditions show the same finer cloud sampled at sub-monthly resolution, so that a month is visible as the average of a restless swarm rather than a single state.¹⁷ Neither condition draws the reconstructed profile: linking the plane back to the column would give the experimental group information the control lacks.

Fidelity. Both conditions read the same latent coordinates. The sign of each principal component is arbitrary, fixed only by the reconstruction basis, and the two conditions render the plane in identical orientation, so that the cloud a control participant sees is the same cloud, not its mirror. The plane preserves equal units on both axes, so that distance on screen is distance in the latent space.

Where the model enters. Here the model is not substrate but ground. The plane the viewer moves through does not exist in nature: it is the basis the decomposition found, and every position in it is a learned projection. Scene I used the latent coordinates to animate a physical object; Scene II makes the coordinate system itself the object of attention, which is the step from expert analytics to public communication that §7.1 claims. The recurrence a participant is asked to look for is a property of that learned space, and it is perceptible only because H2 established the space is flat enough to be walked without distortion. The anomaly agitation is again the IsolationForest, and the sound again takes pc_1 as input; but unlike Scene I, nothing here would exist without the model.

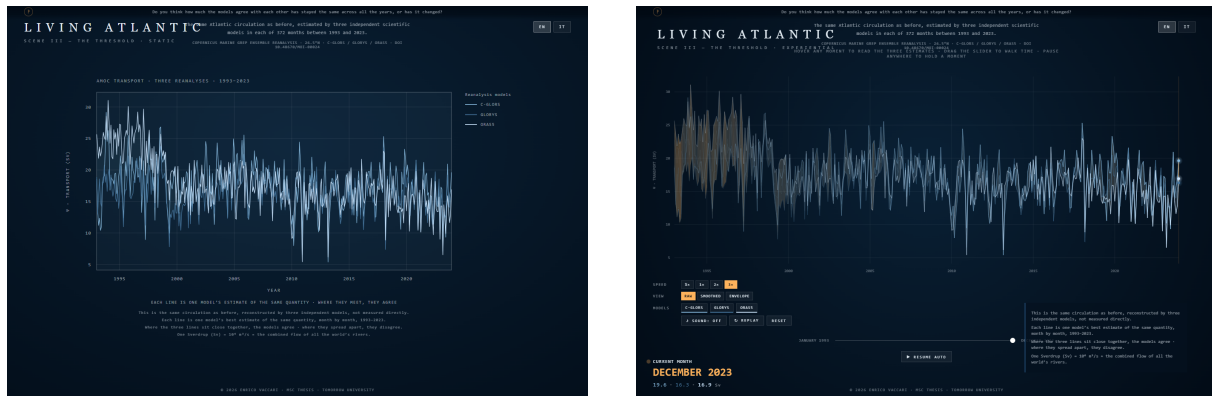
Expected contribution. The static scatter shows that the Atlantic occupies a bounded range of states. The experiential condition shows what the scatter contains but cannot display: that the system moves through that range with structure, drifting at some times and lurching at others, and that it comes back.

7.3.3 Scene III: The Threshold

Driving Question. *Do you think how much the models agree with one another has stayed the same across all the years, or has it changed?*

¹⁶ The three largest steps fall on months the detect phase independently flags as anomalous — an internal consistency recorded in Appendix C, not a claim made on screen.

¹⁷ From `manifold_cloud` in the bundle. Anomaly colouring is taken from the IsolationForest score of each point's month, not from distance to the cloud centroid; a spatial surrogate would encode "far from centre" as "rare", which is part of the reading the scene must withhold.



(a) Control (C1): the static representation of the threshold state.

(b) Experiential (C2+C3): the evolving representation of uncertainty beyond the threshold.

Figure 7.3: The two administered conditions of Scene III, generated from the same data bundle through the same reconstruction function. The static condition presents the threshold structure as a fixed representation, while the experiential condition reveals the divergence and evolution of possible trajectories.

Goal. To make uncertainty a felt quality rather than a number. Scenes I and II showed a quantity measured by an array in the water; Scene III shows the same quantity reconstructed three different ways, by three ocean reanalyses that agree in the present and disagree in the past.¹⁸ The driving question asks whether how much the models agree with one another has stayed the same across the years or has changed.

Representational strategy. The canonical idiom plots the ensemble as a mean with a shaded band, or as member lines overlaid. Either contains the changing disagreement but presents it as a distance to be measured between curves — the inferential task Scene I sets out to remove. Scene III instead makes disagreement a property of the medium: where the reconstructions converge the display is calm, where they diverge it becomes visibly and audibly unsettled. This realises P4, the principle held in reserve through the first two scenes, and with it design proposition DP2.2: the transition from measurement to reconstruction is rendered as perceptual change rather than numerical annotation.

Mapping. P0 gives the strongest channel to the most important quantity, which here is not the transport value but the changing disagreement, and assigns it to motion. P1 assigns each member's estimate to position against time. P2 assigns the passage of time to real motion. P4 assigns uncertainty to animated variability: turbulence, band width and colour temperature all scale with the real spread between members at each month, and none of them is ever printed as a figure. P5 assigns the same spread to audio texture, clean when the models agree and roughening as they part. P3 is absent here as anomaly is not central to the claim.

Conditions. C1 draws the three member lines against time, distinguishable and named, static and unannotated: no era is shaded, no year marked. C2+C3 sets the same lines in motion and adds the living medium between them, with controls that let a viewer re-view identical data at different aggregations —

¹⁸ Copernicus Marine GREP ensemble reanalysis, members C-GLORS, GLORYS and ORAS5, monthly from January 1993 to December 2023.

raw, smoothed over a chosen window, or the envelope alone — and isolate any single reanalysis.¹⁹

Fidelity. Every visual and auditory property is driven by the real spread between the three members at each month, taken from the bundle. No agitation is invented for effect, and no aggregation introduces a quantity the members do not contain. The three lines are never reduced to a mean with error bars: the scene’s subject is three disagreeing reconstructions, not one estimate with a margin.

Where the model enters. Least of the three, and the asymmetry is instructive. The three reanalyses are physical ocean models produced by other institutions, not learned representations built for this thesis, and the spread between them is arithmetic. The machine-learning contribution of Chapter ?? does not drive this scene. What carries across is the grammar rather than the model: the same principles that mapped latent structure to motion in Scenes I and II here map ensemble disagreement to animated variability, and they do so without a latent space to stand on. That the grammar survives the removal of the model is the strongest evidence available within this thesis for objective O2.2, which claims it as transferable beyond this case.

Expected contribution. The static plot shows that three models estimate the same quantity and do not fully agree. The experiential condition shows that the degree of their agreement is itself a variable: the reconstructions part company in the earlier record and converge as observational constraint improves, the mean spread roughly halving between the pre-2004 decade and the years the array has been in the water.²⁰ Whether a non-specialist reads that change as a change in certainty — rather than as a decorative variation — is what expectation E3.2 tests.

Table 7.1: The three scenes of *Living Atlantic*: the question each puts to the viewer, the principles it realises, and what the machine learning contributes. The contribution decreases across the descent—and the grammar’s survival into Scene III, where no learned model drives the display, is the evidence for its transferability (O2.2).

Scene	Driving question	Principles	What ML adds
Scene I The Column	Does the water move all in the same way; if not, in which directions?	P0, P1, P2, P3, P5	The latent coordinates the profile moves along, and the anomaly scores that disturb its texture. The particle field is classical.
Scene II The Engine Room	Does the ocean always move somewhere new, or does it pass back through where it has been?	P1, P2, P3, P5	The plane itself: the space walked is the learned basis. Nothing in the scene exists without the model.
Scene III The Threshold	Has the models’ agreement stayed the same across the years, or changed?	P0, P1, P2, P4, P5	Least of the three. The reanalyses are external physical models; the grammar carries the scene without a learned space.

¹⁹ These are views of one dataset, not additional information: every aggregation is computed from the same three member series, and the raw values remain available in both conditions.

²⁰ The narrowing is gradual rather than a step at the array’s installation, and the artefact marks no date; the change is left to be perceived rather than announced.

Part III

EVALUATE

Understanding the Impact of Representation

Chapter 8

User Study

[To be drafted in Week 7. The qualitative exploratory study: participant recruitment, think-aloud protocol, ethical considerations, analytical framework (thematic analysis).]

Addressed: *Testing · Survey · Think-aloud protocol*

Chapter 9

Results

[To be drafted in Week 8–9. Analytical results from Part I (already obtained), design results from Part II (the realised prototypes), empirical results from Part III (the user study findings).]

Chapter 10

Discussion

[To be drafted in Week 9. Interpretation of the analytical null results (rigour as contribution), the design propositions as realised artefacts, the user-study findings in conversation with the literature on cognition and climate communication, and reflection on the limitations of the methodology.]

Chapter 11

Conclusion and Future Work

[To be drafted in Week 9. Restatement of the contribution, honest scope claims, explicit future work (LSTM forecasting, extension to other tipping elements such as the Greenland Ice Sheet as a fully analytical case study).]

11.1 Reproducibility and Open Science

The scientific claims made in this thesis are grounded in analyses that can be reproduced in full from open sources. The RAPID observational data used throughout Part II is publicly distributed by the British Oceanographic Data Centre (Ben I Moat et al., 2026); the Copernicus Marine Service reanalysis ensemble is accessible under the Copernicus data licence. The analytical pipeline that produced every figure and quantitative result reported in Chapter 6 is maintained as a public repository, with a fixed conda environment specification, versioned notebooks and a documented workflow: <https://github.com/enricovaccari/amoc-thesis-pipeline>. The writing of this thesis, including its L^AT_EX sources and its bibliographic database, is maintained as a separate public repository at <https://github.com/enricovaccari/amoc-thesis-writing>. The commitment underlying this openness is that a scientific claim which cannot be independently verified is, in a meaningful sense, not a scientific claim; the technical documentation required to reproduce the results reported here is provided in full in Appendix C.

Chapter 12

Reflection on Leadership & Personal Growth

12.1 Leadership Journey

12.2 Inspirational Leadership

12.3 Communicating Impactfully

12.4 Alignment with Careers Vectors

Appendix A

Ethics, Boundaries and Limitations

This appendix consolidates, in one place, the methodological boundaries within which the conclusions of this thesis should be read. It is organised around the three phases of the detect–translate–evaluate arc. Each entry states a boundary, the reason it was chosen, and — where relevant — the direction in which future work could extend beyond it. The intent is transparency: to make the limits of the work legible rather than to disperse them across the main text.

A.1 Detect — Analytical Phase

Pattern detection versus prediction

The analytical phase recovers latent structure and characterises anomalies; it does not forecast. The Early Warning Signals analysis returns a null result once the effective sample size is corrected for the autocorrelation of overlapping windows: an apparently strong monotonic trend is shown to be an artefact of pseudo-replication. This is reported as a contribution to methodological honesty, not as evidence for or against an impending transition. No claim about the future state of the AMOC is made or implied by the analytical phase; the deferred forecasting layer (O1.4) is specification only.

Observation versus reconstruction

The analysis draws on two sources of different epistemic status. The RAPID array is direct observation; the Copernicus GREP ensemble is a reanalysis — model output constrained by data assimilation. Claims resting on the reanalysis carry the additional dependence of a model, and are framed accordingly throughout. The two are never conflated: treating a reconstructed value as if it were observed would misrepresent the evidential basis of the analysis, and is avoided by design.

Depth sampling

The RAPID vertical grid is near-uniform (spacing 19.33–19.87 m, a max/min ratio of 1.03), not denser in the upper ocean. No layer-thickness weighting is therefore applied in the decomposition: with levels this evenly spaced, weighting would scale every level by an almost identical constant, without effect on the loadings or variance shares. This is a deliberate, documented choice rather than an oversight.

A.2 Translate — Design Phase

The fidelity constraint on design

Every value in the artefact derives from the analytical bundle, never from an invented scaling. This keeps the representation faithful rather than illustrative, at a deliberate cost: aesthetics are constrained to what

the data support, and where an artistic choice would conflict with the data, the data prevail.

Transferability of the design grammar

The design grammar maps system properties to perceptual channels through documented cognitive principles. Its transferability is a claim, not a proof: the principles are general, but their faithful application depends on the specific structure a system exposes. A mapping that is faithful for the AMOC's ordered, low-dimensional structure may not transfer unchanged to a system with different structure. The grammar is offered as a reusable starting point subject to domain adaptation, not as a fixed template.

A.3 Evaluate — Empirical Phase

Design of the comparison

The evaluation compares two groups of non-specialist participants: a control group exposed to the conventional static condition (C1) and an experimental group exposed to the dynamic and experiential conditions (C2 and C3). The comparison is therefore between two whole representational experiences rather than a within-subject progression through all three conditions. This choice removes the order and repeated-exposure confound that a sequential C1→C2→C3 design would introduce between groups, and it aligns the study directly with the central research question, which contrasts conventional static formats with dynamic and experiential ones.

Form versus amount of exposure

Because the experimental group experiences C2 and C3 while the control group experiences only C1, the groups differ in the extent of exposure as well as its kind, and the design does not isolate the effect of experiential form from the effect of richer exposure. It compares two representational experiences as wholes, which is exactly what the research question asks. Relatedly, the experimental block does not isolate the contribution of interactivity (C2) from that of machine-learning-driven experientiality (C3); a minor within-block order effect (C2 precedes C3) is acknowledged and does not bear on the between-group comparison. Separating these contributions is left to future work. These are properties of the design, acknowledged rather than corrected.

Asynchronous, self-guided delivery

The study is delivered as a self-guided online questionnaire, without the researcher present. This choice suits an evaluation of non-specialists in their own setting, but it shapes what the evidence can be: it captures the mental model a participant expresses in writing, not the reasoning process as it unfolds moment to moment. The evidence is therefore the articulated outcome rather than the live process, and is interpreted as such.

Bilingual instrument

The questionnaire is administered in Italian and in English, so participants respond in their first language. Open responses are analysed in the language in which they are written, rather than translated before coding, so that translation does not distort meaning. The thematic analysis therefore spans two languages, and themes are reconciled across them at the reporting stage.

Evidential weight and sample size

The study is exploratory. Its evidential weight rests on the qualitative analysis of how participants articulate their understanding in open written responses; any closed comprehension items are reported descriptively, not as inferential tests, consistent with the sample size (approximately 24–30 participants, evenly split) and the exploratory framing. The study does not claim statistical inference to a wider population.

Interpretation of the two-group comparison

The expectations guiding the empirical phase (E3.1, E3.2) are framed around the between-group contrast and do not presuppose the direction or magnitude of any effect. Qualitative differences in how the two groups articulate their understanding may appear with or without accompanying differences in the descriptive comprehension items; both outcomes are reported. The study is not designed to adjudicate whether such differences would persist at population scale.

The unassessed closing sequence

After the assessed part of the questionnaire, every participant is shown the fuller experiential sequence once, for interest. Responses to it are not assessed and do not enter the comparison. It functions as a debrief and a return to the participant — so that those who saw only the static version are not left without the experience — and is placed outside the assessed journey by design, so that it cannot act as a confound.

A.4 Ethical Conduct of the Study

Participation is informed and voluntary. Each participant receives a plain-language information sheet explaining what the study involves, what data are collected and how they are used, and gives consent before taking part; they may stop at any point without giving a reason and without consequence. No deception is used. The topic is not sensitive — participants reason about an ocean current, not about anything personal or distressing — so the principal ethical duties are the standard ones of consent, anonymity and data protection. Responses are anonymised, with identifying details removed or replaced by codes, so that no participant is identifiable in the thesis or any output. Data are stored securely and in line with the General Data Protection Regulation, kept only as long as the research requires, and accessible only to the researcher and, where necessary, the supervisor. The study proceeds only once the supervisor has confirmed that the appropriate ethical approval or exemption for a study of this scale is in place. The full information sheet, consent text and data-management detail are reproduced in Appendix B.

Appendix B

Study Protocols

This appendix documents the instruments of the empirical evaluation. The study is delivered as a self-guided online questionnaire, so the protocol is the questionnaire itself: its structure, its four variants, its comprehension items, and the consent and information text that precede it. Where the exact wording is still being finalised as part of the work in progress, this is marked explicitly; the structure around it is fixed.

B.1 Questionnaire Variants

The instrument exists in four variants, crossing two languages with two conditions:

- **Language:** Italian and English, so that participants respond in their first language. Open responses are analysed in the language in which they are written (Appendix A.3).
- **Condition:** a *control* version showing only the conventional static representation (C1), and an *experimental* version showing the dynamic and experiential representation (C2 and C3).

The four variants (IT-control, IT-experimental, EN-control, EN-experimental) are identical in structure, background questions, and comprehension items. They differ only in language and in which representational sequences the participant is shown. This keeps the two groups comparable: the information conveyed is held constant, and only the representational form varies (Appendix A.3).

B.2 Structure of the Questionnaire

Each participant follows the same journey, in four stages.

1. Consent and information. The questionnaire opens with the plain-language information sheet and the consent statement (§B.4). Participation proceeds only after consent is given.

2. Background questions. A short set of questions characterises the participant and allows prior knowledge to be accounted for in analysis:

- age band;
- occupation or field of work or study;
- prior familiarity with the AMOC or with ocean-climate systems (a short self-rated scale, with an option to describe any prior knowledge).

These items collect no directly identifying information; responses are anonymised (Appendix A.4).

3. Guided exposure with interleaved comprehension. The participant is taken through the representational sequences of their assigned condition. Comprehension questions are placed *between* sequences rather than only at the end, so that understanding is captured as it forms. The comprehension items are described in §B.3.

4. Unassessed closing sequence. After the assessed part is complete, every participant — including the control group — is shown the fuller experiential sequence once, for interest. Responses to it are not assessed and do not enter the between-group comparison; it serves as a debrief and a return to the participant, and is placed outside the assessed journey by design (Appendix A.3).

B.3 Comprehension Items

The comprehension items are the study's primary evidence. They target the three properties the representation is designed to convey — the system's dynamics, its non-linearity, and its uncertainty — so that they measure understanding of what the artefact actually encodes rather than general climate knowledge.

The items lean on *open-ended* questions, which reveal a participant's mental model rather than their ability to recognise a supplied answer, with *closed* items possibly added in support to give a signal directly comparable between the two groups. The open responses are the primary material for the reflexive thematic analysis; any closed items are reported descriptively (Appendix A.3). The exact wording, number, and open/closed balance are finalised close to data collection, drafted and reviewed with the supervisor, and piloted before the study proper.

[TO BE FINALISED: the frozen set of comprehension items, in both languages, with their placement between sequences. Draft items target: (i) how the system changes over time; (ii) whether its response is proportional or can shift abruptly; (iii) how confident one can be in what is shown.]

B.4 Information Sheet and Consent

[TO BE FINALISED: the full participant information sheet and consent statement, in both languages. The information sheet states, in plain language: what the study involves and how long it takes; that participation is voluntary and may be stopped at any point without consequence; what data are collected (background questions and questionnaire responses) and that no directly identifying information is gathered; how responses are anonymised, stored securely under the GDPR, retained only as long as needed, and used only for this research; and contact details for the researcher and supervisor. Consent is given explicitly before the questionnaire begins.]

The ethical commitments these documents implement are set out in Appendix A.4.

Appendix C

Code and Reproducibility

This appendix documents the code behind the thesis, so that the analytical pipeline and the representational artefact can be inspected and reproduced. It covers two repositories, the computational environment, the steps to reproduce the analysis, and a provenance ledger that traces every perceivable element of the artefact back to its source in the data, the design principle that licenses it, and its computational origin. The guiding commitment is the one that runs through the whole thesis: every value shown derives from the data, and the path from data to display is open to inspection.

C.1 Repositories

The work is held in two research repositories. The first contains the analytical pipeline — the code that turns the observational and reanalysis records into the latent structure the artefact is built on. The second contains the representational artefact — the browser-based code that renders that structure as the three conditions of the study.¹

Analytical pipeline

Repository: [FILL: amoc-thesis-pipeline] — [FILL:GitHubURL].

This repository contains the notebooks and modules that implement the *detect* phase. The analysis is organised as a sequence of numbered notebooks, each a self-contained stage whose output feeds the next:

- `01_data_first_look` — loading and inspection of the RAPID record;
- `02_rapid_vs_copernicus` — comparison of observation against the reanalysis ensemble;
- `03_PCA_on_vertical_profiles` — the linear decomposition of the vertical structure;
- `04_anomalies_early_warning_signals` — anomaly detection and the corrected early-warning analysis;
- `05_autoencoder_vertical_profiles` — the non-linear model comparison that tests manifold flatness;
- `06_export_for_translate` — assembly of the data bundle consumed by the artefact.

Shared functionality — data loading, paths, common transforms — is factored into supporting modules [FILL: e.g. `data_loading.py`, `paths.py`] rather than duplicated across notebooks. The final notebook exports a single JSON bundle that is the one source of truth for every value the artefact displays (§C.3).

¹ A third repository holds the written thesis ($\LaTeX$ sources). It is not documented here: it is not a research contribution and is not needed to reproduce the analysis or the artefact.

Representational artefact

Repository: [FILL: repo name] — [FILL:GitHubURL].

This repository contains the browser-based artefact: the three representational conditions built for the study, rendered with WebGL (through Three.js) and the Web Audio API. Its structure separates the three scenes from the shared machinery they depend on:

- the scene files, one static and one experiential per scene [FILL: confirm file names];
- a shared module layer [FILL: e.g. bundle.js, reconstruct.js, palette.js] that loads the data bundle and reconstructs the profiles, so that informational equivalence across conditions is guaranteed by shared provenance rather than asserted — one bundle, one reconstruction function, shared imports, not duplicated code.

This shared-module architecture is what lets the thesis claim that the static and experiential conditions carry the same information: they draw on the same bundle and the same reconstruction, and differ only in how they render it.

C.2 Environment and Reproduction

The analytical pipeline runs in Python [FILL: 3.11] in a Conda environment [FILL: amoc-thesis]. The principal libraries are `xarray` and `netCDF4` for the ocean data, `numpy`, `pandas` and `scipy` for numerical work, `scikit-learn` for PCA and the Isolation Forest, `statsmodels` for the time-series statistics, and [FILL: e.g. PyTorch / TensorFlow] for the autoencoder. Plotting uses `matplotlib` and `seaborn`. The exact versions are pinned in the repository’s environment file [FILL: environment.yml / requirements.txt].

To reproduce the analysis, the environment is recreated from that file and the numbered notebooks are run in order; the data sources (RAPID and the Copernicus GREP ensemble) are obtained from their providers [FILL: note access - DOIs / download instructions] and placed as described in the repository README. Running the notebooks in sequence regenerates the data bundle, from which the artefact can then be served locally in a browser [FILL: any serve step, e.g. a local static server].

C.3 Provenance Ledger

The provenance ledger (Table C.1) is the reproducibility claim made concrete. For every perceivable element of the artefact, it names the field of the data bundle it derives from, the design-grammar principle that licenses its perceptual encoding, and its computational origin — whether empirical, classically derived, or produced by machine learning. Nothing perceivable is omitted except non-informational scaffolding (colours, fonts, tints), which carries no data. The ledger is what turns the fidelity principle from a promise into something a reader can audit line by line.

Table C.1: Provenance ledger for the three scenes. Every perceivable element, its source in the data bundle, the design principle that licenses it, and its computational origin. Colours, fonts and tints are non-informational scaffolding and are omitted.

Scene	Element	Bundle field	Prin.	Origin
I	$\Psi(z)$ curve, static (C1)	mean_profile via Eq. 7.1, pc = 0	P1	Empirical (record mean)
I	$\Psi(z)$ curve, animated (C2/C3)	mean_profile + trajectory_monthly	P1, P2	ML (latent trajectory; H2)
I	Particle field	$\partial\Psi/\partial z$ of current profile	P2	Classical (derivative)
I	Anomaly texture	anomalies.clusters (score)	P3	ML (IsolationForest)
I	Sound	trajectory_monthly.pc1	P5	ML input, classical synth.
I	Depth / peak readout	depth_m; max of profile	P1	Derived
II	Monthly points	trajectory_monthly.pc1/P2	P1, P2	ML (latent coordinates)
II	Path motion	trajectory_monthly or- der	P2	ML (latent trajectory)
II	Sub-monthly cloud	manifold_cloud.pc	P1	ML (latent coordinates)
II	Anomaly agitation	anomalies.clusters (score)	P3	ML (IsolationForest)
II	Equilibrium glow	centroid of manifold_cloud	P1	Derived (mean)
II	Sound	trajectory_monthly.pc1	P5	ML input, classical synth.
III	Three member lines	copernicus_ensemble.members	P4	External reanalyses
III	Living medium	upper – lower per month	P4	Classical (spread)
III	Colour temperature	upper – lower per month	P4	Classical (spread)
III	Aggregation views	members (rolling mean, en- velope)	P1	Derived
III	Sound texture	upper – lower per month	P5	Classical, classical synth.

References

- [Cover art]. (2009). *2009 13th International Conference Information Visualisation*, C1–C1. <https://doi.org/10.1109/IV.2009.114>
- Armstrong McKay, D. I., Staal, A., Abrams, J. F., Winkelmann, R., Sakschewski, B., Loriani, S., Fetzer, I., Cornell, S. E., Rockström, J., & Lenton, T. M. (2022). Exceeding 1.5°C global warming could trigger multiple climate tipping points. *Science*, 377(6611), eabn7950. <https://doi.org/10.1126/science.abn7950>
- Artificial Botany*. (2019–present). fuse*. Retrieved July 13, 2026, from <https://www.fuseworks.it/series/artificial-botany>
- Baldi, P., & Hornik, K. (1989). Neural networks and principal component analysis: Learning from examples without local minima. *Neural Networks*, 2(1), 53–58. [https://doi.org/10.1016/0893-6080\(89\)90014-2](https://doi.org/10.1016/0893-6080(89)90014-2)
- Ben I Moat, David Smeed, Darren Rayner, William E Johns, Ryan H Smith, Denis L Volkov, Shane Elipot, Tillys Petit, Jules B Kajtar, Molly O Baringer, & Julie Collins. (2026). *Atlantic meridional overturning circulation observed by the RAPID-MOCHA-WBTS (RAPID-Meridional Overturning Circulation and Heatflux Array-Western Boundary Time Series) array at 26N from 2004 to 2024 (v2024.1a)*. (Documents,Network Common Data Form,Text or Plaintext; Version 12). Documents,Network Common Data Form,Text or Plaintext. NERC EDS British Oceanographic Data Centre NOC. <https://doi.org/10.5285/48D0BF43-0598-CEB2-E063-7086ABC062F1>
- Ben-Yami, M., Morr, A., Bathiany, S., & Boers, N. (2024). Uncertainties too large to predict tipping times of major Earth system components from historical data. *Science Advances*, 10(31), eadl4841. <https://doi.org/10.1126/sciadv.adl4841>
- Bertin, J., Berg, W. J., & Bertin, J. (1983). *Semiology of graphics: Diagrams, networks, maps* (1. ed). ESRI Press.
- Boers, N. (2021). Observation-based early-warning signals for a collapse of the Atlantic Meridional Overturning Circulation. *Nature Climate Change*, 11(8), 680–688. <https://doi.org/10.1038/s41558-021-01097-4>
- Böning, C. W., Behrens, E., Biastoch, A., Getzlaff, K., & Bamber, J. L. (2016). Emerging impact of Greenland meltwater on deepwater formation in the North Atlantic Ocean. *Nature Geoscience*, 9(7), 523–527. <https://doi.org/10.1038/ngeo2740>
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp063oa>
- Braun, V., & Clarke, V. (2019). Reflecting on reflexive thematic analysis. *Qualitative Research in Sport, Exercise and Health*, 11(4), 589–597. <https://doi.org/10.1080/2159676X.2019.1628806>
- Buckley, M. W., & Marshall, J. (2016). Observations, inferences, and mechanisms of the Atlantic Meridional Overturning Circulation: A review. *Reviews of Geophysics*, 54(1), 5–63. <https://doi.org/10.1002/2015RG000493>
- Cabrera, Á. A., Perer, A., & Hong, J. I. (2023). Improving Human-AI Collaboration With Descriptions of AI Behavior. *Proceedings of the ACM on Human-Computer Interaction*, 7, 1–21. <https://doi.org/10.1145/3579612>
- Caesar, L., Rahmstorf, S., Robinson, A., Feulner, G., & Saba, V. (2018). Observed fingerprint of a weakening Atlantic Ocean overturning circulation. *Nature*, 556(7700), 191–196. Retrieved May 21, 2026, from <https://mural.maynoothuniversity.ie/id/eprint/13174/>

- Camps-Valls, G., Tuia, D., Zhu, X. X., & Reichstein, M. (Eds.). (2021, September 27). *Deep Learning for the Earth Sciences: A Comprehensive Approach to Remote Sensing, Climate Science, and Geosciences* (1st ed.). Wiley. <https://doi.org/10.1002/9781119646181>
- Cleveland, W. S., & McGill, R. (1984). Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods. *Journal of the American Statistical Association*, 79(387), 531–554. <https://doi.org/10.1080/01621459.1984.10478080>
- Creative, V. (n.d.). *Climate Visuals for COP22 and beyond*. Retrieved July 3, 2026, from <https://climateoutreach.org/reports/climate-visuals-briefing-for-cop22/>
- Cronin, M. A., Gonzalez, C., & Sterman, J. D. (2009). Why don't well-educated adults understand accumulation? A challenge to researchers, educators, and citizens. *Organizational Behavior and Human Decision Processes*, 108(1), 116–130. <https://doi.org/10.1016/j.obhdp.2008.03.003>
- Dakos, V., Carpenter, S. R., Brock, W. A., Ellison, A. M., Guttal, V., Ives, A. R., Kéfi, S., Livina, V., Seekell, D. A., Van Nes, E. H., & Scheffer, M. (2012). Methods for Detecting Early Warnings of Critical Transitions in Time Series Illustrated Using Simulated Ecological Data (B. Yener, Ed.). *PLoS ONE*, 7(7), e41010. <https://doi.org/10.1371/journal.pone.0041010>
- Ditlevsen, P., & Ditlevsen, S. (2023). Warning of a forthcoming collapse of the Atlantic meridional overturning circulation. *Nature Communications*, 14(1), 4254. <https://doi.org/10.1038/s41467-023-39810-w>
- European Union-Copernicus Marine Service. (2019). *Global Ocean Ensemble Physics Reanalysis*. Mercator Ocean International. <https://doi.org/10.48670/MOI-00024>
- Fischer, H., Van Den Broek, K. L., Ramisch, K., & Okan, Y. (2020). When IPCC graphs can foster or bias understanding: Evidence among decision-makers from governmental and non-governmental institutions. *Environmental Research Letters*, 15(11), 114041. <https://doi.org/10.1088/1748-9326/abbc3c>
- Frajka-Williams, E., Foukal, N., & Danabasoglu, G. (2023). Should AMOC observations continue: How and why? *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 381(2262), 20220195. <https://doi.org/10.1098/rsta.2022.0195>
- Global Warming's Six Americas, September 2021*. (n.d.). Retrieved July 3, 2026, from <https://climatecommunication.yale.edu/publications/global-warmings-six-americas-september-2021/>
- Hannachi, A., Jolliffe, I. T., & Stephenson, D. B. (2007). Empirical orthogonal functions and related techniques in atmospheric science: A review. *International Journal of Climatology*, 27(9), 1119–1152. <https://doi.org/10.1002/joc.1499>
- Harold, J., Lorenzoni, I., Shipley, T. F., & Coventry, K. R. (2016). Cognitive and psychological science insights to improve climate change data visualization. *Nature Climate Change*, 6(12), 1080–1089. <https://doi.org/10.1038/nclimate3162>
- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design Science in Information Systems Research1. *MIS Quarterly*, 28(1), 75–106. <https://doi.org/10.2307/25148625>
- Hullman, J., Resnick, P., & Adar, E. (2015). Hypothetical Outcome Plots Outperform Error Bars and Violin Plots for Inferences about Reliability of Variable Ordering. *PLOS ONE*, 10(11), e0142444. <https://doi.org/10.1371/journal.pone.0142444>
- Iten, R., Metger, T., Wilming, H., Del Rio, L., & Renner, R. (2020). Discovering Physical Concepts with Neural Networks. *Physical Review Letters*, 124(1), 010508. <https://doi.org/10.1103/PhysRevLett.124.010508>
- Jansen, Y., Dragicevic, P., Isenberg, P., Alexander, J., Karnik, A., Kildal, J., Subramanian, S., & Hornbæk, K. (2015). Opportunities and Challenges for Data Physicalization. *Proceedings of*

- the 33rd Annual ACM Conference on Human Factors in Computing Systems, 3227–3236. <https://doi.org/10.1145/2702123.2702180>
- Kahan, D. M. (2015). Climate-Science Communication and the *Measurement Problem*. *Political Psychology*, 36(S1), 1–43. <https://doi.org/10.1111/pops.12244>
- Kale, A., Nguyen, F., Kay, M., & Hullman, J. (2019). Hypothetical Outcome Plots Help Untrained Observers Judge Trends in Ambiguous Data. *IEEE Transactions on Visualization and Computer Graphics*, 25(1), 892–902. <https://doi.org/10.1109/TVCG.2018.2864909>
- Kause, A., Bruine de Bruin, W., Domingos, S., Mittal, N., Lowe, J., & Fung, F. (2020). Communications about uncertainty in scientific climate-related findings: A qualitative systematic review. *Environmental Research Letters*, 16(5), 053005. <https://doi.org/10.1088/1748-9326/abb265>
- Kause, A., Bruine de Bruin, W., Millward-Hopkins, J., & Olsson, H. (2021). Public perceptions of how to reduce carbon footprints of consumer food choices. *Environmental Research Letters*, 14(11), 114005. <https://doi.org/10.1088/1748-9326/ab465d>
- Kellstedt, P. M., Zahran, S., & Vedlitz, A. (2008). Personal Efficacy, the Information Environment, and Attitudes Toward Global Warming and Climate Change in the United States. *Risk Analysis*, 28(1), 113–126. <https://doi.org/10.1111/j.1539-6924.2008.01010.x>
- Kirsh, D. (2013). Embodied cognition and the magical future of interaction design. *ACM Transactions on Computer-Human Interaction*, 20(1), 1–30. <https://doi.org/10.1145/2442106.2442109>
- Lakoff, G. (1999). *Philosophy In The Flesh*.
- Lang, C. (2014). Do weather fluctuations cause people to seek information about climate change? *Climatic Change*, 125(3–4), 291–303. <https://doi.org/10.1007/s10584-014-1180-6>
- Lenton, T. M., Held, H., Kriegler, E., Hall, J. W., Lucht, W., Rahmstorf, S., & Schellnhuber, H. J. (2008). Tipping elements in the Earth’s climate system. *Proceedings of the National Academy of Sciences*, 105(6), 1786–1793. <https://doi.org/10.1073/pnas.0705414105>
- Marlon, J. R., Bloodhart, B., Ballew, M. T., Rolfe-Redding, J., Roser-Renouf, C., Leiserowitz, A., & Maibach, E. (2019). How Hope and Doubt Affect Climate Change Mobilization. *Frontiers in Communication*, 4, 20. <https://doi.org/10.3389/fcomm.2019.00020>
- Marriott, K., Schreiber, F., Dwyer, T., Klein, K., Riche, N. H., Itoh, T., Stuerzlinger, W., & Thomas, B. H. (Eds.). (2018). *Immersive Analytics* (Vol. 11190). Springer International Publishing. <https://doi.org/10.1007/978-3-030-01388-2>
- Morris, B. S., Chrysochou, P., Christensen, J. D., Orquin, J. L., Barraza, J., Zak, P. J., & Mitkidis, P. (2019). Stories vs. facts: Triggering emotion and action-taking on climate change. *Climatic Change*, 154(1–2), 19–36. <https://doi.org/10.1007/s10584-019-02425-6>
- Nicholson-Cole, S. A. (2005). Representing climate change futures: A critique on the use of images for visual communication. *Computers, Environment and Urban Systems*, 29(3), 255–273. <https://doi.org/10.1016/j.compenvurbsys.2004.05.002>
- Ojala, M. (2023). Hope and climate-change engagement from a psychological perspective. *Current Opinion in Psychology*, 49, 101514. <https://doi.org/10.1016/j.copsyc.2022.101514>
- O’Neill, S. J., & Smith, N. (2014). Climate change and visual imagery. *WIREs Climate Change*, 5(1), 73–87. <https://doi.org/10.1002/wcc.249>
- Padilla, L. M., Creem-Regehr, S. H., Hegarty, M., & Stefanucci, J. K. (2018). Decision making with visualizations: A cognitive framework across disciplines. *Cognitive Research: Principles and Implications*, 3(1), 29. <https://doi.org/10.1186/s41235-018-0120-9>

- Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A Design Science Research Methodology for Information Systems Research. *Journal of Management Information Systems*, 24(3), 45–77. <https://doi.org/10.2753/MIS0742-1222240302>
- PrintMag. (2017, January 30). *Data Humanism: The Revolutionary Future of Data Visualization*. Retrieved July 6, 2026, from <https://www.printmag.com/article/data-humanism-future-of-data-visualization/>
- Rahmstorf, S., Box, J. E., Feulner, G., Mann, M. E., Robinson, A., Rutherford, S., & Schaffernicht, E. J. (2015). Exceptional twentieth-century slowdown in Atlantic Ocean overturning circulation. *Nature Climate Change*, 5(5), 475–480. <https://doi.org/10.1038/nclimate2554>
- Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., & Prabhat. (2019). Deep learning and process understanding for data-driven Earth system science. *Nature*, 566(7743), 195–204. <https://doi.org/10.1038/s41586-019-0912-1>
- Sacha, D., Zhang, L., Sedlmair, M., Lee, J. A., Peltonen, J., Weiskopf, D., North, S. C., & Keim, D. A. (2017a, January). *Visual Interaction with Dimensionality Reduction: A Structured Literature Analysis* (Vol. 23). <https://doi.org/10.1109/TVCG.2016.2598495>
- Sacha, D., Zhang, L., Sedlmair, M., Lee, J. A., Peltonen, J., Weiskopf, D., North, S. C., & Keim, D. A. (2017b). Visual Interaction with Dimensionality Reduction: A Structured Literature Analysis. *IEEE Transactions on Visualization and Computer Graphics*, 23(1), 241–250. <https://doi.org/10.1109/TVCG.2016.2598495>
- Scheffer, M., Bascompte, J., Brock, W. A., Brovkin, V., Carpenter, S. R., Dakos, V., Held, H., Van Nes, E. H., Rietkerk, M., & Sugihara, G. (2009). Early-warning signals for critical transitions. *Nature*, 461(7260), 53–59. <https://doi.org/10.1038/nature08227>
- Segel, E., & Heer, J. (2010). Narrative Visualization: Telling Stories with Data. *IEEE Transactions on Visualization and Computer Graphics*, 16(6), 1139–1148. <https://doi.org/10.1109/TVCG.2010.179>
- Sheppard, S. R. (2015). Making climate change visible: A critical role for landscape professionals. *Landscape and Urban Planning*, 142, 95–105. <https://doi.org/10.1016/j.landurbplan.2015.07.006>
- Smeed, D. A., Josey, S. A., Beaulieu, C., Johns, W. E., Moat, B. I., Frajka-Williams, E., Rayner, D., Meinen, C. S., Baringer, M. O., Bryden, H. L., & McCarthy, G. D. (2018). The North Atlantic Ocean Is in a State of Reduced Overturning. *Geophysical Research Letters*, 45(3), 1527–1533. <https://doi.org/10.1002/2017GL076350>
- The Sonification Handbook* | edited by Hermann, Hunt, Neuhoff. (n.d.). Retrieved July 2, 2026, from <https://sonification.de/handbook/>
- Sonnewald, M., Lguensat, R., Jones, D. C., Dueben, P. D., Brajard, J., & Balaji, V. (2021). Bridging observations, theory and numerical simulation of the ocean using machine learning. *Environmental Research Letters*, 16(7), 073008. <https://doi.org/10.1088/1748-9326/ac0eb0>
- Sonnewald, M., Wunsch, C., & Heimbach, P. (2019). Unsupervised Learning Reveals Geography of Global Ocean Dynamical Regions. *Earth and Space Science*, 6(5), 784–794. <https://doi.org/10.1029/2018EA000519>
- Spiegelhalter, D. (2017). Risk and Uncertainty Communication. *Annual Review of Statistics and Its Application*, 4(1), 31–60. <https://doi.org/10.1146/annurev-statistics-010814-020148>
- Sterman, J. D. (2008). Risk Communication on Climate: Mental Models and Mass Balance. *Science*, 322(5901), 532–533. <https://doi.org/10.1126/science.1162574>
- Stoknes, P. E. (2015). *What We Think About When We Try Not to Think About Global Warming: Toward a New Psychology of Climate Action*. Retrieved July 1, 2026, from https://www.researchgate.net/publication/280683963_What_We_Think_About_When_We_Try_Not_To_Think_About_Global_Warming

- Stommel, H. (1961). Thermohaline Convection with Two Stable Regimes of Flow. *Tellus*, 13(2), 224–230. <https://doi.org/10.1111/j.2153-3490.1961.tb00079.x>
- Sweller, J. (1988). Cognitive Load During Problem Solving: Effects on Learning. *Cognitive Science*, 12(2), 257–285. https://doi.org/10.1207/s15516709cog1202_4
- Sweller, J., Van Merriënboer, J. J. G., & Paas, F. (2019). Cognitive Architecture and Instructional Design: 20 Years Later. *Educational Psychology Review*, 31(2), 261–292. <https://doi.org/10.1007/s10648-019-09465-5>
- Transforming our world: The 2030 Agenda for Sustainable Development* | Department of Economic and Social Affairs. (n.d.). Retrieved July 2, 2026, from <https://sdgs.un.org/2030agenda>
- Tversky, B., Morrison, J. B., & Betrancourt, M. (2002). Animation: Can it facilitate? *International Journal of Human-Computer Studies*, 57(4), 247–262. <https://doi.org/10.1006/ijhc.2002.1017>
- Tversky, B. G. (2019). *Mind in motion: How action shapes thought*. Basic Books.
- Van Der Linden, S. (2015). The social-psychological determinants of climate change risk perceptions: Towards a comprehensive model. *Journal of Environmental Psychology*, 41, 112–124. <https://doi.org/10.1016/j.jenvp.2014.11.012>
- Ware, C. (2021). *Information visualization: Perception for design* (Fourth edition). Morgan Kaufmann, Inc.